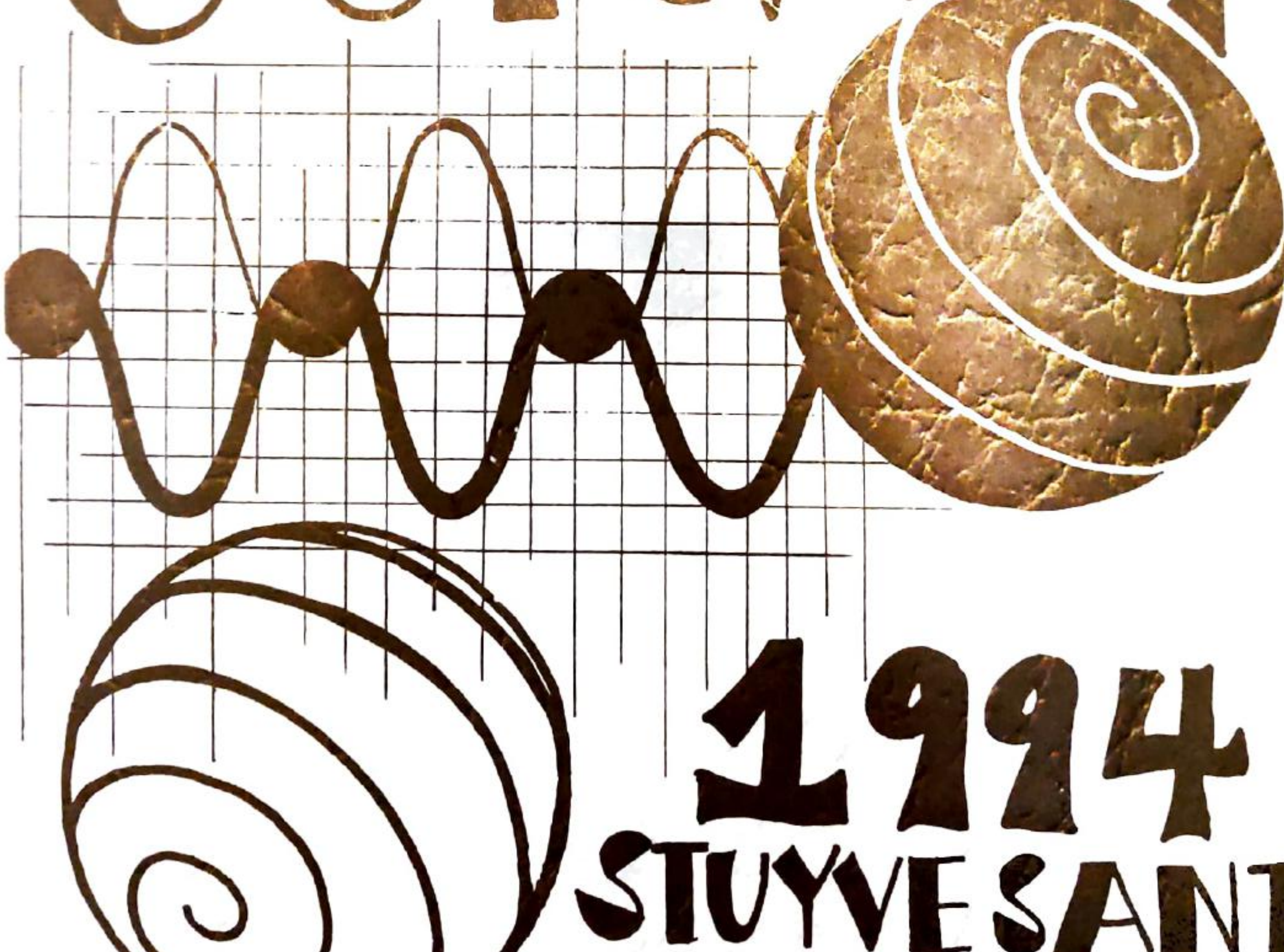
MATH SURVEY



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Department of Mathematics
Stuyvesant High School
345 Chambers Street
New York, New York 10282
212-312-4800

MATH SURVEY

Table of Contents

A Note From Iris and Helen, the Editors:

We have had an interesting time organizing this year's *Math Survey*. We have enjoyed working with many of our new staff members and have appreciated the numerous articles that were submitted for publication.

As editors-in-chief, we would like to thank our staff members for their dedication and hard work. A sincere and special thanks goes out to the faculty members who have offered us support and access to certain school facilities: Dr. Rothenberg and Mr. Clancy for your advice and counsel and Mr. Bellush and Ms. Shimmel for your office space and use. We couldn't have finished our job without your generosity. Lastly, we are grateful for the support we received from you, our buyers, readers, and advertisers.

We hope that you enjoy reading *Math Survey* as much as much as we have enjoyed providing it for you.

Sincerely,

Helen J. Hong
Iris Lan

Helen Hong
Iris Lan,
Editors-in-chief

How to Stabilize the Solutions of Linear Algebraic Equations with Gaussian Elimination

BY IRIS LAN

ABSTRACT

The accurate and stable solutions are the most essential for linear system equations. This paper will analyze how to find stable solutions through the proving and scaling methods.

I. INTRODUCTION

A set of n linear simultaneous equations

$$
\begin{gathered}
a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}X_n = b_1 \\
a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}X_n = b_2 \\
a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \ldots + a_{3n}X_n = b_3 \\
\vdots \\
a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \ldots + a_{mn}X_n = b_m
\end{gathered}
$$

may be written in matrix form as **A * X = B** where A is an n x n coefficient matrix and **X** and **b** are column vectors of the variables and right-hand-sides respectively. Gaussian elimination methods are the most important methods of solving simulaneous equations. In the analysis of the Gaussian Elimination algorithm, analysts have extensively studied stability and instability. In order to control error in the computed solution of a linear system, pivoting, scaling, and the condition number will be explored. In this paper, the error analysis will approach a more complete understanding of this algorithm.

II. PIVOTING STRATEGIES

The number $a_{r,r}$ in the position *(r,r)* that is used to eliminate x_r in the rows $r+1,\ r+2,\ \ldots\ ,\ N$ is called the *r*th pivotal element and the *r*th row is called the pivotal row. We now introduce three pivoting strategies.

1. PP (Partial Pivoting):
   Choosing the largest nonzero entry in the pivot column as the pivot element.
2. CP (Complete Pivoting):
   Choosing the largest nonzero entry in the entire augmented matrix as the pivot.
3. PR (Rook Pivoting):
   Confines the search for the kth pivot to rows k through n.

Example:

Given a linear system A * X = B, the augmented matrix form can be defined as:

$$
\begin{bmatrix} A & B \end{bmatrix} = \begin{bmatrix}
4 & 4{,}800 & -3{,}200 & 1{,}600 & -11{,}200 \\
-1 & -1{,}196 & 1{,}700 & 1{,}200 & 8{,}400 \\
2 & 2{,}401 & -1{,}373 & 7{,}600 & -9800 \\
1 & 1{,}199 & -1{,}024 & 4{,}102 & -700 \\
-2 & -2{,}398 & 2{,}049 & -4{,}699 & 14{,}357 \\
-9{,}596 & 8{,}903 & -2{,}570 & 3{,}478 & 11{,}358
\end{bmatrix}
$$

The completed solutions by using used PP, CP,and RP:

The concept of number is the obvious distinction between the beast and man. Thanks to number, the cry becomes

$$X_j$$

j	Partial Pivoting	Complete Pivoting	Rook Pivoting
1	1.00	0.99	1.00
2	1.00	1.00	0.99
3	1.00	0.99	1.00
4	1.00	1.00	0.99
5	1.14	1.14	1.14

III. SCALING

Defining the scale of each row:

$$s_i = \max_{1\le j\le n} |a_{ij}| \; ; \; s_i \ne 0.$$

The appropriate row interchange to place zeros in the first column is determined by choosing the least integer k with

$$\frac{|a_{k1}|}{s_k} = \mathrm{MAX}_{1\le j\le n} \frac{|z_{j1}|}{s_j}.$$

The effect of scaling to ensure that the largest element in each row has a relative magnitude one before the comparsion for row interchange is performed. The scaling is done only for comparison purpose, so the division by scaling factors produces no round-off error in the system. The scaling system does not alter the oreintation of the linear systems.

Example:

Given linear algebraic equations with order 2:

$$\begin{bmatrix} A & B \end{bmatrix} = \begin{bmatrix} 3 & 97 & 500 \\ 0.0029 & -0.0001 & 0.014 \end{bmatrix},$$

the exact solution is

$$X = \begin{bmatrix} 5.00 & 5.00 \end{bmatrix}'.$$

Using the power of ten, the elements are of about the same size.
Then the above systems can be changed to read:

$$\begin{bmatrix} A & B \end{bmatrix} = \begin{bmatrix} 3 & 97 & 500 \\ 29 & -1 & 140 \end{bmatrix}.$$

Then the solution is

$$X = \begin{bmatrix} 5.00 & 5.01 \end{bmatrix}'.$$

The scaling could lead to a better computer for a given linear system.

IV. CONDITION NUMBERS

Let A be a matrix, and let X be a vector. The definition of the condition number is

$$\|A\|_2 \|A^{-1}\|_2$$

where

$$\|A\|_2 = \frac{\underset{X \neq 0}{\mathrm{MAX}} \|AX\|_2}{\|X\|}$$

Example:

There is a full linear system of order 3. Then

$$[A \quad B] = \begin{bmatrix} 9{,}700 & 9{,}500 & -2{,}000 & 950{,}000 \\ 970 & 980 & 950 & 14{,}500 \\ 970 & 979 & -21 & 9{,}640 \end{bmatrix}.$$

The condition number is approximately 6550. After using the row-scaled system, we can get the the following matrix:

$$[A \quad B] = \begin{bmatrix} 97 & 95 & -2 & 950 \\ 97 & 98 & 95 & 1{,}450 \\ 97 & 97.9 & -2.1 & 964 \end{bmatrix}.$$

The condition number is approximately 168.

Therefore, the size of the condition number is a measure of how good a solution we can expect from the direct methods.

SUMMARY

Pivoting and scaling can influence solutions. We can choose from the above stable ways to get more accurate solutions.

REFERENCES

G. Pool and L. Neal, *Gaussian Elimination: When is Scaling Benefical?*, Linear Algebraic Appl., 162-164:309-324, 1992.

G. Pool and L. Neal, *A Geometric Analysis of Gaussian Elimination I*, Linear Algebraic Appl., 149:249-272, 1991.

G. Pool and L. Neal, *A Geometric Analysis of Gaussian Elimination II*, Linear Algebraic Appl., 173:239-264, 1992.

L. N. Trefethen and R. S. Schreiber, *Average-case Stability of Gaussian Elimniation*, SIAM J. Matrix Anal. Appl., 1:335-360,1990.

Lee W. Johnson and R. Dean Riess, *Numerical Analysis*.

J. H. Wilkinson, *Modern Error Analysis*, SIAM Rev., 13:548-568, 1971.

Extensions of the Ellipse to the "Trillipse"

BY SAUL BLUMENTHAL

CONIC SECTIONS MAY be defined as curves resulting from the intersection of a right circular cone and a plane. If the plane doesn't pass through the cone's vertex (in which case, a degenerate conic - c point, line, or two intersecting lines - results), one of four conics may be formed. They are the circle, ellipse, parabola and hyperbola. Any conic section may be described by the general second degree equation :

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

Each conic section may also be defined as the locus (set) of all points that satisfy certain geometric conditions. For example, the ellipse may be described as the locus of points the sum of whose distances from two fixed points called foci remains constant. If the foci are at $F_1(-a,0)$ and $F_2(a,0)$, with $a>0$, and the constant sum is $2S$, with $S>0$, then this is equivalent to $\{P(x,y){:}d(P,F_1)+d(P,F_2)=2S\}$. (The notation $\{P(x,y){:}[\text{condition}]\}$ represents the locus of all points that satisfy the condition, and the notation $d(A,B)$ refers to the distance from point A to point B). (By using the distance formula, the condition of the ellipse is: $\sqrt{(x+a)^2+y^2}+\sqrt{(x-a)^2+y^2}=2S$. After some algebraic manipulation, this becomes $\frac{x^2}{S^2}+\frac{y^2}{S^2-a^2}=1$. One question that arises from this locus definition of the ellipse is as follows : "What is the locus that results from having the same condition as the ellipse, but with three, instead of two, foci ?" This locus shall be called the "trillipse," and is defined as follows: $\{P(x,y){:}d(P,F_1)+d(P,F_2)+d(P,F_3)=k\}$, where k is a constant. The trillipse to be investigated will have a constant sum $2T$, and foci at $F_1(-a,0)$, $F_2(a,0)$ and $F_3(0,0)$, with $a>0$, and $T>0$. The definition of this locus may be represented as follows: $\sqrt{(x+a)^2+y^2}+\sqrt{x^2+y^2}+\sqrt{(x-a)^2+y^2}=2T$. However, because of the presence of three radicals, this equation is not easily simplified as was the case with the equation of the ellipse. Another strategy is therefore necessary. The method to be explored involves finding expressions for x and y, each in terms of certain parameters that represent distances to specific points. Because the equations for the trillipse build on those for the ellipse, the first step was to represent the ellipse parametrically. This is done as follows.

Let $P(x,y)$ be on the ellipse with the given specifications, and let d be its distance from the focus $F_1(-a,0)$. In order to satisfy the condition of the ellipse, it must therefore be a distance $2S-d$ from the focus $F_2(a,0)$; that is, $d(P,F_1)+d(P,F_2)=d+(2S-d)=2S$. This may be represented by the following equivalency :

$$\{P(x,y){:}d(P,F_1(-a,0))+d(P,F_2(a,0))=2S\}\Leftrightarrow$$
$$\{P(x,y){:}d(P,F_1(-a,0))=d \text{ and } d(P,F_2(a,0))=2S-d\}.$$

This latter expression represents geometrically the intersection of a circle of radius d centered at $(-a,0)$, and a circle of radius $2S-d$ centered at $(a,0)$; these points of intersection may be found by solving the system of the two equations of these circles:

$$(x+a)^2+y^2=d^2 \leftrightarrow x^2+2ax+a^2+y^2=d^2$$
$$(x-a)^2+y^2=(2S-d)^2 \leftrightarrow x^2-2ax+a^2+y^2=4S^2-4Sd+d^2.$$

Subtracting the second equation on the right from the first yields :

Angling may be said to be so like the mathematics that it can never be fully learnt. — Izaak Walton

$$4ax = 4Sd - 4S^2 = 4S(d - S) \leftrightarrow x = \frac{S}{a}(d - S)$$

From the equation of the first circle, $y^2 = d^2 - (x + a)^2$, and substitution for x we get the following:

$$y = \pm\sqrt{\left(1 - \frac{S^2}{a^2}\right)(d - [S + a])(d - [S - a])}$$

From the triangle inequality, $d(P, F_1(-a,0)) + d(P, F_2(a,0)) = 2S > 2a \leftrightarrow S > a$, and therefore, in order for the expression under the radical in the equation for y to be non-negative, it is found that $S - a \le d \le S + a$.

This technique may also be used to attempt to find equations for the trillipse with the previously mentioned specifications. We again let $P(x,y)$ be on the trillipse, and let e be its distance from $F_3(0,0)$. In order that $d(P,F_1) + d(P,F_2) + d(P,F_3) = 2T$, we have $d(P,F_1) + d(P,F_2) + e = 2T$ and $d(P,F_1) + d(P,F_2) = 2T - e$. This latter expression represents an ellipse with foci at $F_1(-a,0)$ and $F_2(a,0)$ with a constant sum $2T$-e, and the points on the trillipse may be found by the intersection of this ellipse with a circle or radius e centered at $F_3(0,0)$. This is represented by the following equivalency:

$$\{P(x,y){:}d(P,F_1(-a,0) + d(P,F_2(a,0)) + d(P,F_3(0,0)) = 2T\} \Leftrightarrow$$

$$\{P(x,y){:}d(P,F_3(0,0)) = e \text{ and } d(P,F_1(-a,0)) + d(P,F_2(a,0)) = 2T - e\}.$$

In other words, the trillipse is generated by the intersection of an expanding circle with a simultaneously contracting ellipse, which is analogous to how the ellipse is generated.

Using the derived equations for the ellipse with $2S = 2T - e \leftrightarrow S = T - \frac{e}{2}$ we have

$$x = \frac{T - \frac{e}{2}}{a}\left(d - \left[T - \frac{e}{2}\right]\right) \text{ and}$$

$$y = \pm\sqrt{\left(1 - \frac{\left(T - \frac{e}{2}\right)^2}{a^2}\right)\left(d - \left[\left(T - \frac{e}{2}\right) + a\right]\right)\left(d - \left[\left(T - \frac{e}{2}\right) - a\right]\right)}.$$

We also know that from the definition of e, $x^2 + y^2 = e^2$. Substitution of the above expression for x and y in this last equation results in $d^2 - 2\left(T - \frac{e}{2}\right)d + \left(-\frac{e^2}{2} - 2Te + 2T^2 - a^2\right) = 0$ and by the quadratic formula, we have the following equation for d in terms of e : $d = T - \frac{e}{2} \pm \sqrt{\frac{3}{4}e^2 + Te - T^2 + a^2}$.

Substitution into the expression for x results in : $x = \pm\frac{T - \frac{e}{2}}{a}\sqrt{\frac{3}{4}e^2 + Te - T^2 + a^2}$.

Substitution into the expression for y, however, yields an equation which seemingly is not as easily simplified as that for x, so another approach to the problem was attempted.

In finding parametric equations for the ellipse, we can also use as the parameter the distance from $P(x,y)$ to the origin (0,0), instead of to the focus $F_1(-a,0)$. Let this distance be f. We wish to express the distance from $P(x,y)$ to the point $F_1(-a,0)$ in terms of f. This distance, by the distance formula, is $\sqrt{(x+a)^2+y^2}=\sqrt{x^2+2ax+a^2+y^2}$. However, since $x^2+y^2=f^2$, we have that $d\left(P,F_1(-a,0)\right)=\sqrt{f^2+2ax+a^2}$. If this expression is used instead of d in the presented derivation of the parametric equation for the ellipse, the following results: $x=\frac{S}{a}\left(\sqrt{f^2+2ax+a^2}-S\right)$. We wi sh to solve for x in terms of f, which after some algebraic manipulation, yields: $x=\pm\frac{S}{a}\sqrt{f^2+a^2-S^2}$. Since $x^2+y^2=f^2$, we substitute for x and solve for y, yielding : $y=\pm\sqrt{\left(1-\frac{S^2}{a^3}\right)\left(f^2-S^2\right)}$. Because of the radicals in the expressions for x and y, it is found that the following restriction exist on f: $\sqrt{S^2-a^3}\le f\le S$.

If one replaces the sum $2S$ by $2T-f\left(S=T-\frac{f}{2}\right)$, then for a given f, $d(P,F_1(-a,0))+d(P,F_2(a,0))=2T-f$. However, since f = $d(P,(0,0))$, then we have $d(P,(-a,0))+d(P,(a,0))+d(P,(0,0))=(2T-f)+f=2T$, and therefore, $P(x,y)$ satisfies the condition of the trillipse. This new set of parametric equation is as follows:

$$x=\frac{T-\frac{f}{2}}{a}\sqrt{f^2+a^2-\left(T-\frac{f}{3}\right)^2}$$

$$y=\pm\sqrt{\left(1-\frac{\left(T-\frac{f}{2}\right)^2}{a^2}\right)\left(f^2-\left(T-\frac{f}{2}\right)^2\right)}.$$

By expanding the expression for x, we obtain $x=\frac{T-\frac{f}{2}}{a}\sqrt{\frac{3}{4}f^2+Tf-T^2+a^2}$ which was exactly what was found using the first approach, if e is replaced by f. The equation for y is the simplification for which we were searching with the first approach upon substitution in y, but the transformation of that expression into this one is not at all obvious.

The main advantage of using this parametric representation of the trillipse is that it connects the ellipse with the trillipse. This derivation tells us that the points on a trillipse are points on an ellipse with foci at $(-a,0)$ and $(a,0)$ and with a "varying constant sum."

As with the ellipse, certain restrictions exist on the parameter f, which are found to be as follows :

If $a<T<\frac{3}{2}a$ then $\frac{-2+\sqrt{4T^2-3a^2}}{3}\le f\le 2(T-a)$ and if $T\ge\frac{3}{2}a$ then $\frac{-2+\sqrt{4T^2-3a^2}}{3}\le f\le\frac{2}{3}T$.

Using the equations that have been derived, functions were defined for the computer program MATLAB v3.5 (IBM). Various graphs of the ellipse and trillipse were then obtained. After obtaining the graph of the trillipse with specification $2T=50$, $a=10$, it appeared that the trillipse might be an ellipse. This was shown not to be true, however, by examining the ellipse with the same x- and y-intercepts as the given trillipse. It was found that for a given x value, the corresponding y values for the trillipse and ellipse differed by very small amounts (e.g. 0.06096); this suffices, though, to illustrate that the general trillipse is not an ellipse, since it was shown to be true for a specific case. Although it has not been formally proven, it may be seen from the graphs that the trillipse is also not a parabola or hyperbola, that is, the trillipse is not a conic section.

Using the techniques developed herein, a more general trillipse, with foci at $F_1(-a,0)$, $F_2(a,0)$ and $F_3(h,k)$, for any h and k, may be investigated. Even more general is the "n-llipse," the locus of points the sum of whose distances from n fixed foci is constant. These foci would be at $F_1(-a,0)$, $F_2(a,0)$, $F_3(h_3,k_3)$, . . . , $F_n(h_n,k_n)$.

As an example, an expression for x for the trillipse with foci at $F_1(-a,0)$, $F_2(a,0)$ and $F_3(b,0)$, for any b, and constant sum $2T$ was found to be : $x = \dfrac{T-\frac{g}{2}}{a}\left(\dfrac{T-\frac{g}{2}}{a}b \pm \sqrt{g^2+a^2-\left(T-\frac{g}{2}\right)^2+\left(\dfrac{\left(T-\frac{g}{2}\right)^2}{a^2}-1\right)b^2}\right)$.

Besides by increasing the number of foci, there is another way in which the ellipse may be generalized, through changing the relationship between the distances to the foci. For example, if instead of keeping the sum $d(P,F_1)+d(P,F_2)$ constant, the difference $d(P,F_1)-d(P,F_2)$ is kept constant. Then the curve generated forms one branch of a hyperbola. Constant products and quotients may also be considered. These relationships may also be applied to loci with any number of foci. The general locus is described as follows:

$\{P(x,y): F(d(P,F_1)+d(P,F_2),...,d(P,F_n))=k\}$, where $F_1,F_2,...,F_n$ are the foci, F is any function prescribing a relationship between the given distances, and k is a constant. These loci may be investigated through the use of techniques similar to those employed for the trillipse.

One aspect of the trillipse which has not been considered is its differentiability; it appears from the graphs that there are singularities (points where the derivative does not exist) for the case of $T=\frac{3}{2}a$ at $(-a,0)$ and $(a,0)$. One may investigate under what circumstances such singularities might exist for the trillipse as well as for the n-llipse. Definitions of the n-llipse analogous to the plane-cone definition of conic sections may also be explored.

There are some possible applications of this work, which include 1) situations in which it is desirable to maintain certain distances from given fixed locations, such as in military campaigns, airplane routing and mail delivery, and 2) a better understanding of the shapes of planetary orbits, which are not exactly elliptical because of the presence of many complex interacting forces.

In summary, although parametric equations were only found for a specific case of the trillipse, it was shown how equations may be found for the general trillipse and for n-llipse with more than three foci, using similar techniques, that is, this research has laid the foundation for the study of what is perhaps a new class of curves.

Bibliography

Burton, William Wilder. *Analytic Geometry*. New York: Harcourt Brace and Company, 1938.

Coolidge, Julian Lowell. *A History of the Conic Sections and Quadric Surfaces*. New York: Dover Publications, Inc., 1968.

Larson, Roland E. and Hostetler, Robert P. *Calculus and Analytic Geometry (Alternate Third Edition)*. Massachussetts: DC Heath and Company, 1986.

Salmon, George. *Treatise on Conic Sections (Sixth Edition)*. New York: Chelsea Publishing Company.

Thomas, George B., Jr. *Calculus and Analytic Geometry*. Massachussetts: Addison-Wesley Publishing Company, Inc., 1954.

I admit that twice two makes four is an excellent thing, but if we are to give everything its due, twice two makes five is sometimes a very charming thing too. — Feodor Dostoievski (1821-1881)

Hilbert's Tenth Problem

BY ERIC CHO

IN AUGUST 8, 1900, David Hilbert delivered a keynote address to the Second International Congress of Mathematicians in Paris, and in his speech he presented a list of twenty-three major unsolved problems to challenge future mathematicians. "We hear within us the perpetual call: this is the problem," he said. "Seek its solution. You can find it by pure reason, for in mathematics there is no *ignorabimus*." Among the 23 problems, one has become quite famous: Hilbert's Tenth Problem.

The problem is this: To devise an algorithm to determine whether a Diophantine equation with any number of unknown quantities has a solution. A Diophantine equation is any equation with one or more variables that has integer coefficients and integer solutions. For example, if $x^2 + y^2 = 2$ is treated as a regular equation for real numbers, then there are infinitely many solutions. However, if it is treated as a Diophantine equation, then there are only four solutions:

$$1)\ x=1,\ y=1;\ 2)\ x=-1,\ y=1;\ 3)\ x=1,\ y=-1;\ \text{and } 4)\ x=-1,\ y=-1.$$

If the equation is changed to $x^2 + y^2 = 3$, then there are still infinitely many real solutions. For a Diophantine equation in this form, however, there are no solutions.

A few methods do exist to determine the solvability of some types of Diophantine equations. The linear equation in the form $ax = b$ has an integer solution if, and only if, a divides b exactly. The equation $ax + by = c$ has an integer solution if the greatest common factor d, of a and b, divides c exactly. For second-degree equations with one or two unknowns, such as $x^2 + 5x - 6 = 0$ or $x^2 + 2xy - y^2 = 3$, one can use Gauss' quadratic reciprocity theory. Hilbert, however, asks for a *general* algorithm applicable to every Diophantine equation.

Does such an algorithm exist? Hilbert had also said in his address: "Occasionally it happens that we seek the solution under insufficient hypothesis or in an incorrect sense, and for this reason do not succeed. The problem then arises: to show impossibility of the solution under the given hypotheses, or in the sense contemplated." This is precisely what the Russian mathematician Yuri Matyasevich did in 1970, when he proved that Hilbert's algorithm indeed does not exist.

Computability

A set is said to be *computable* if there is an algorithmic method of determining which numbers belong to the set and which do not. Therefore, a set of integers S is said to be computable if, for example, one can construct a computer program with a finite number of steps which, given any integer, stops with an output of 1 if the integer is a member of S and stops with an output of 0 if the integer is not a member of S.

A set of integers S can be said to be *listable*, or recursively enumerable, if there is a program which, given any integer as input and after a finite number of steps, stops with an output of 1 if and only if the integer is a member of S. Thus, the program indicates only whether the integer is a member of the set, not whether it isn't.

If $\overline{S}$ denotes the complement of S, that is, the set of all integers that are not members of S, then it can be proven that if S is computable, then both S and $\overline{S}$ are listable. If S is computable, then the program that indicates this fact will also indicate that S is listable (since the program will stop with an output of 1 if the input is a member of S). By adding a final step that switches the outputs of 0 and 1, the program will also indicate that $\overline{S}$ is listable.

The converse is also true. If S and $\overline{S}$ are listable, then S is computable. Let P and Q be programs which indicate the listability of S and $\overline{S}$, respectively. A program U is then constructed such that it runs P and Q

Mathematics possesses not only truth, but supreme beauty—a beauty cold and austere, like that of sculpture.
— Bertrand Russell

alternately on a given input until one of these stops with output 1. If P stops with an output of 1, then U outputs 1. If Q stops with an output of 1, then U outputs 0. The program U, in effect, determines the computability of S. Thus, if S and $\overline{S}$ are listable, then S is computable because there exists an algorithm to determine whether or not a given value is a member of the set.

The crucial idea in the solution of Hilbert's 10th Problem is this: that there is a set K that is listable but not computable. To illustrate this fact, let U be a program consisting of programs $P_1, P_2, P_3, \ldots$, and let K be the set of all natural numbers r such that P_r halts with output 1 when U receives r as an input (note that this condition makes K listable). To show that K is not computable, first *assume that it is computable*. Then this means that $\overline{K}$ is listable and that there is some program P_s which indicates this. Since $\overline{K}$ is the complement of K, P_s should halt with an output of 1 for any input, say for 150, if and only if P_{150} does not halt with output 1 for 150. Thus P_s is not the same as P_{150}. Similarly, P_s is not the same as P_r for any other value of r. The same applies to any other number, not just 150, showing that no program P_s exists for $\overline{K}$. Therefore, $\overline{K}$ is not listable and K is not computable.

Matyasevich's Solution

The first attempt to prove the nonexistence of Hilbert's algorithm was made by Martin Davis in 1950. His strategy was to prove that for every listable set S there is a corresponding polynomial $Q_s(x, y_1, y_2, \ldots, y_n)$, with integer coefficients, such that a positive integer k is a member of S if and only if the Diophantine equation $Q_s(x, y_1, y_2, \ldots, y_n) = 0$ has a solution. It was later determined that the degree of Q_s need not be greater than 4 and the number of variables need not exceed 14. It is not known yet, however, if both of these bounds can be achieved at the same time.

If it were true that every listable set S has an associated polynomial Q_s then this would imply that Hilbert's algorithm to determine the solvability of Diophantine equations is nonexistent. To illustrate this fact, assume on the contrary that such an algorithm exists. Let S be a listable but non-computable set of integers, as previously described in the section on computability. Assuming that Hilbert's algorithm exists would imply that there also exists a program P which stops with output 1 on input k if the Diophantine equation $Q_s(k, y_1, y_2, \ldots, y_n) = 0$ has a solution, and stops with output 0 if Q_s does not have a solution. Because of the relationship between S and Q_s, however, P would then indicate that S is computable. This contradicts the original condition that S be listable but non-computable. The only way out of this dilemma is to conclude that no such algorithm exists.

Davis, however, was unable to prove that the polynomial Q_s always exists for every listable set S. The answer turned out to lie in Julia Robinson's own investigation of sets that can be defined by Diophantine equations. She developed various techniques for handling equations whose solutions behaved in an exponential manner. In 1960, Robinson, Davis and Hilary Putnam worked together in proving that if even one Diophantine equation could be found whose solutions behaved exponentially in an appropriate sense, then it would be possible to describe every listable set by a Diophantine equation. This would prove that Hilbert's algorithm does not exist. Unfortunately, the three mathematicians were in turn frustrated in their search for such an equation.

This last step in the proof was finally completed in 1970 by Matyasevich. He found such an equation by making use of what are known as Fibonacci numbers. The importance of the sequence lay in the fact that the Fibonacci numbers grow exponentially: the n^{th} number in the sequence is approximately equal to

$$\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n.$$

If it were possible to find a Diophantine equation whose solutions were approximately related to the Fibonacci numbers, then the Davis-Robinson-Putnam condition would be satisfied and the solution to Hilbert's 10th Problem would be complete.

Matyasevich obtained such an equation by squaring each of the following ten polynomial equations and then adding them together to obtain one large and complicated equation:

1) $u+w-v-2=0$	6) $m-c(2h+g)-3=0$
2) $l-2v-2a-1=0$	7) $m-fl-2=0$
3) $l^2-lz-z^2-1=0$	8) $x^2-mxy+y^2-1=0$
4) $g-bl^2=0$	9) $(d-1)l+u-x-1=0$
5) $g^2-gh-h^2-1=0$	10) $x-v-(2h+g)(l-1)=0$

In these equations the values of u and v are related in such a way that v is the $(2u)$th Fibonacci number, which is enough to satisfy the condition of the Davis-Robinson-Putnam result. Therefore, for every listable set there is an associated Diophantine equation, and since there exists a listable set K that is not computable no algorithm can exist which tests the solvability of a Diophantine equation.

Thus, Hilbert's 10th Problem was finally conquered.

Bibliography

Davis, Martin and Hersh, Ruben. "Hilbert's Tenth Problem," *Scientific American*, Nov. 1973, pp. 84-91.
Hopcroft, John E. "Turing Machines," *Scientific American*, May 1984, pp. 86-98.
Stewart, Ian. *The Problems of Mathematics*. Great Britain: Oxford University Press, 1987, pp. 213-221.

THE MISSING NUMBER

Supply the missing digit.

THE CHINESE are wonderfully expert at figures. The professor in the illustration asked me to write down any two numbers, provided that in forming them I used only the nine digits and zero. For example, I could write:

342195
6087

Each digit was to be used once and once only. I was then told to add the two numbers. Finally, I was asked to erase the two numbers as well as any digit I pleased in the answer. The professor glanced at the answer and quickly told me the missing digit.

The slate in the picture shows my answer. Can you supply the missing digit and explain how the Chinese mathematician was able to guess it so quickly?

Answer

The nine digits add up to 45 which is a multiple of nine. Regardless of how these digits and zero are arranged to make two numbers, the sum of the two numbers will also be a multiple of nine.

Moreover, when you add the digits in any multiple of nine, the result is always a multiple of nine. So we have only to add the digits we see in the answer to obtain 10, then subtract this from 18 (the nearest multiple of nine that is greater than 10) to obtain 8, the missing digit.

Foundations of Mathematics

BY FELIX LIU

The Greeks used to believe that every number is either a whole number or a fraction. It was discovered around the fifth century B.C. that there are numbers that are incommensurable. The Greeks were very upset by the discovery, which contradicted their belief, and they kept it a secret for a while, but eventually they accepted this new concept that has come to be known as irrational numbers. That was the first crisis in the foundations of mathematics.

The second crisis in the foundations of mathematics came more than two thousands years later during the nineteenth century. The problem was the belief that continuity implies differentiability. Fortunately, by the end of the century, French and German mathematicians solved that problem. A new subject, known as the theory of functions, was developed as a result.

The third crisis in the foundations of mathematics, which has not yet ended, began late in the nineteenth century, following the creation of set theory by Georg Cantor. In this article, I shall discuss the basic ideas behind logic and set theory. I shall also look at the problems that still exist in the foundations of mathematics.

Right after the creation of set theory, many paradoxes were discovered. But, they were all ignored, mostly because they were quite technical and thus considered insignificant. Early this century, Bertrand Russell discovered a paradox so elementary that most mathematicians were finally convinced that set theory needed to be modified. It eventually led to the axiomatization of set theory.

Consider the set of all sets that do not contain themselves, or equivalently, the set A such that for all B, $B \in A$ if and only if B satisfies the property $\phi(B) = "B \notin B"$. Then, consider whether or not $A \in A$. Suppose $A \in A$. Then $\phi(A)$ is not true, so $A \notin A$. But in that case, $\phi(A)$ is true, and so $A \in A$. This is what is known as Russell's paradox.

Frank P. Ramsey has classified paradoxes into two categories: logical and semantical. Russell's paradox belongs to the former. The most famous example of a paradox of the latter category would probably be the "liar" paradox. Suppose a person says that he is a liar. Is he really one? If he is, then he has just told the truth. If he is not, then he has just lied.

The difference between logical and semantical paradox is that a logical paradox is constructed with mathematical language while a semantical paradox is constructed with our verbal language. Closely related is the concept of the object language and the metalanguage of mathematics. In books about computer programming, we have two types of languages: one is the collection of commands that the computer can understand, such as PRINT or GOTO in BASIC, and the other is the text about these commands. Likewise, object language consists of statements of theories while metalanguage tells something about the theories themselves. For example, "$1 < 2$" is a statement of the object language expressing a relation between the numbers 1 and 2, while "'$1 < 2$' is a true statement" is a statement of metalanguage about a statement of the object language.

The predicate calculus, which is a system of logic, has an object language. The language is such that one can always determine if a statement is a formula of the language. The predicate calculus also has a set of axioms, also known as logical axioms. A proof is a sequence $A_1, A_2, \ldots, A_n$ of formulae such that for all i, A_i is either a logical axiom or inferred from two formulae A_j and A_k, where $j, k < i$ by *Modus Ponen*. A theorem is the last formula in a proof.

The idea of a first-order theory is not too different. First, there must be a way to decide whether a statement is a formula of the language of a first-order theory T. Secondly, there must be a set of axioms of T. A proof in T is the same as one in predicate calculus, except in addition to the logical axioms, the axioms of T can also appear. A theorem is still the last step of a proof.

There are many examples of first-order theory. Number theory, which consists of the five postulates of

The science of pure mathematics, in its modern developments, may claim to be the most original creation of the human spirit. — Alfred North Whitehead

Giuseppe Peano, is a well-known example of a first-order theory. Zermelo-Fraenkel (ZF) set theory, created by Ernst Zermelo and modified by Abraham A. Fraenkel, is another example.

There are some properties that we would like a first-order theory T to have. For example, we would A_j and A_k, where $j,k < i$ want there to be no paradox that can be developed from T; i.e., we want there to be no proof of $A \wedge \sim A$ for any A. A theory with this property is said to be consistent. Also, if A is an arbitrary statement in the object language of T, we would want it to be possible to prove it or disprove it. A theory with such a property is said to be complete. If T is not complete, and A is a statement that can neither be proved nor be disproved from T, then A is said to be independent of T.

At the beginning of the century, David Hilbert led a group of mathematicians in trying to develop a theory T such that the consistency and completeness of T can be proved in T, and that all the important mathematical results known to us can be derived in T. He never succeeded, and Kurt Gödel showed the reason in two theorems, now known as Gödel's first and second incompleteness theorems. The first incompleteness theorem states that the number theory with Peano's axiom, or any extension of it, is incomplete if the system is consistent. The second incompleteness theorem states that the consistency of number theory, or any extension of it, cannot be proved in the system.

While a theory with all the properties that Hilbert wanted doesn't exist, there are theories with some of the properties. The Zermelo-Fraenkel set theory is one such theory. ZF is not a complete theory, but most of the results in mathematics can be developed from it, and although one cannot prove in ZF its consistency, there are good reasons to believe that it is consistent.

One can derive mathematics from set theory mainly because one can construct the set of real numbers from it. We shall not go into the construction, however. Another important area in set theory centers upon the idea of cardinality. Two sets A and B have equal cardinality, or are equipollent, if there exists a bijection of A onto B. For different reasons, it is desired to have a representative set for each cardinality. These representatives are cardinal numbers. Usually, the theory of cardinal numbers is derived from the theory of ordinal numbers.

After its construction, mathematicians wanted to know if all cardinalities are represented by some cardinal number. The answer to that question is independent of ZF; however, the answer is yes if an axiom, known as the axiom of choice (AC), is added to ZF. AC was a very controversial axiom during the first half of the twentieth century. Today, it is accepted by most people; however, many mathematicians still indicate it when AC is used in the proof of a theorem.

AC was first introduced by Zermelo. It states that if $\{S_i\}$, $i \in A$ is a family of non-empty sets, then there is a function f such that for every $i \in A$, $f(S_i) \in S_i$; i.e., it is possible to choose an element from each of the S_i. The reader may wonder why such an intuitively obvious statement as AC can be controversial. The reason is that it asserts the existence of a set that chooses an element from each set in an *arbitrary* collection of *arbitrary* sets. Remember that Russell's paradox can be derived with the assumption of the existence of a set that doesn't seem to have any problem at the first sight. Also, it has been shown using AC that it is possible to decompose a sphere into parts and reassemble two spheres congruent to the original from the parts, while without AC it's not possible.

AC was not the only thing that presented a great challenge to the mathematicians of the first half of the twentieth century. Mathematicians of the time also found the continuum hypothesis intriguing. The continuum hypothesis states that any infinite set of reals is equipollent to either the set of natural numbers or the set of reals. It is a conjecture that many have tried to prove. There is a generalized version of the continuum hypothesis, known as the Generalized Continuum Hypothesis (GCH).

In 1940, Gödel presented to the mathematical community a very good reason for accepting AC as well as trying to prove GCH. He showed that if AC or GCH is added to ZF, the system we get will be consistent provided that ZF itself is consistent. Of course, it is not possible to prove the consistency of ZFC or ZFC+GCH by Gödel's incompleteness theorem, so Gödel has obtained the best result possible. Seven years after Gödel presented his proof, Waclaw Sierpinski showed that GCH implies AC. The chance of proving the

As far as the laws of mathematics refer to reality, they are not certain, and as far as they are certain, they do not refer to reality. — Albert Einstein continuum hypothesis was shattered in 1963 by a Stuyvesant graduate, Paul J. Cohen, when he showed the independence of AC from ZF and the independence of GCH from ZFC. The method he developed, namely forcing, was later modified by others and used in many other independence proofs. For example, it was shown that it is impossible to construct a non-measurable set, including decomposing a sphere into parts that can be reassembled into two spheres congruent to the original, without using AC.

There are many different schools of thoughts in mathematics. The school of Formalism, led by Hilbert, has developed the foundations of mathematics as I have presented in this article. As the reader may have observed, the Formalist looked at mathematics as a game played on paper with certain rules. The object of the game is to create mathematics without violating these rules. Although the views of the Formalist prevail over all the other schools of thought and is widely accepted as the way the foundations of mathematics should be today, it is not the only view that exists. Other than the school of Formalism, the most successful is probably the school of Intuitionism, which disagrees with the Formalists mainly in the treatment of infinity. The Intuitionists believed that we do not understand infinity adequately enough to use it as we have in the development of modern mathematics. Indeed, our treatment of the link between discrete and continuous objects is not satisfactory, for otherwise, we should know more than we already do about the continuum hypothesis. The relation between discrete and continuous objects is very difficult to understand, and it is considered the central problem in the foundations of mathematics, still to be solved.

I hope the reader now knows more about the foundations of modern mathematics and the problems it faces after having read this article. I have tried my best to avoid getting into technical details so that everyone can understand it. I encourage those who are interested in understanding the details of the development of the foundations of mathematics to read the references cited throughout the article and given below.

References

Banach, S, and Tarski, A., "Sur la décomposition des ensembles de points en parties respectivement congruentes," *Fundamenta Mathematicæ*, Vol.6(1924), 244-277.

Cantor, Georg., "Beiträge zur Begründung der transfiniten Mengenlehre," *Mathematische Annalen*, Vol.46 (1895), 481-512. Translation, *Contributions to the Founding of the Theory of Transfinite Numbers*, 1915.

Cohen, P. J., *Set Theory and the Continuum Hypothesis*. New York, 1966.

Fraenkel, A. A., Bar-Hillel, Y., and Lévy, A., *Foundations of Set Theory*. Amsterdam, 1973.

Gödel, K., "Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I" *Monatshefte für Mathematik und Physik* Vol.38 (1931), 173-198.

Gödel, K., *The Consistency of the Axiom of Choice and the Generalized Continuum Hypothesis with the Axioms of Set Theory*. Princeton, 1940.

Hausdorff, F., *Grundzüge der Mengenlehre*. Leipzig, 1914. Reprinted: New York, 1949.

Jech, T., *Set Theory*. New York, 1978.

Kamke, E, *Theory of Sets*. New York, 1950.

Kleene, S., *Introduction to Metamathematics*. Princeton, 1952.

Kunen, K, *Set Theory: An Introduction to Independence Proofs*. Amsterdam, 1980.

Landau, E., *Grundlagen der Analysis*. Leipzig, 1930. (English edition: *Foundations of Analysis*. New York, 1951.)

Margaris, A., *First Order Mathematical Logic*. Waltham, Massachusetts, 1967.

Russell, B., *Principles of Mathematics*. London, 1903.

Shoenfield, J. R., *Mathematical Logic*. Reading, Massachusetts, 1967.

Sierpinski, W., *Leçons sur les nombres transfinis*. Paris, 1928.

Sierpinski, W., "L'hypothèse généralisé du continu et l'axiome du choix," *Fundamenta Mathematicæ*, Vol.34 (1947), 1-5.

Solovay, R.M., "A model of set theory in which every set of reals is Lebesgue measurable," *Annals of Mathematics*, Vol.92(1970), 1-56.

Histories make men wise; poets, witty; the mathematics, subtile; natural philosophy, deep; moral, grave; logic and

Pairs Prime of Gaussian Integers Differing Linearly

BY LEO VOLFTSUN

IN THIS PAPER, I will find primes which differ linearly in Gaussian integers, offering some nice applications of the original Hardy-Littlewood's Conjecture B, prime density methods and Gaussian integers.

I'll denote the set of all Gaussian integers by G, i.e. G is the subset of the complex numbers consisting of all $x+iy$, where of x and y are both integers. G is a unique factorization domain in which the units are 1, i, -1 and $-i$. If $g = x+iy$ is in G, we define the norm of g, $N(g)$, to be x^2+y^2. We will also use this definition for the complex numbers. The norm of a product is the product of the norms and $N(a)$ divides $N(b)$ if a divides b. The complex number $1+i$ is a prime in G (and a factor of 2) that plays a role in G rather like that played by 2 in the rational integers. (These facts can be found discussed in more detail in **[4]**.)

Before I introduce the first theorem, I'd like to mention the function f in Gaussian integers (in integers the function $f(n)$ gives the number of integers relatively prime to n and less than $f(n)$). In Gaussian integers $f(g)$ is defined as the number of lattice points (points with integer coordinates in Cartesian plane) in the square with coordinates (x,y), $(0,0)$, $(y,-x)$ and $(x+y, y-x)$, where $g = x+iy$. More intuitively, the square described has one of its sides as a segment from (0,0) to (x,y) and lies entirely in the half-plane with positive x-coordinate. Now I am going to establish the formula for $f(g)$, where g is not an integer or a conjugate of an integer (g is such that if $p = a+ib$, then $ab \neq 0$) in 3 lemmas.
Combining these three results, we get the following statement:

Theorem 1: $f(n) = (N(p_1^{a_1}) - N(p_1^{a_1-1})) \times ..... \times (N(p_k^{a_k}) - N(p_k^{a_k-1}))$, where n is a Gaussian integer, such that $n = p_1^{a_1} ..... p_k^{a_k}$, with p_i being a factor of n.

Now I want to introduce the result that has played a major role in prime density related problems. This theorem (then a conjecture) was proposed by Littlewood and Hardy as Conjecture B in their paper. I am going to alter the original statement by changing the system from real to complex numbers, from integers to Gaussian integers, and by substituting $N(x)$ for x.

Theorem 2: There are infinitely many prime pairs $(p, p+m)$ for every even (that is, divisible by $1+i$) m where p and m are Gaussian integers. If $p_m(x)$ is the number of pairs less than $N(x)$, then

$$p_m(x) \sim 2C_i \prod_{p>2,}^{p|m} \frac{p-1}{p-2} \frac{N(x)}{(N(\log x))^2} \quad \text{where } C_i = \prod\nolimits_{p>2} (p - \frac{p}{(p-1)^2}).$$

(p and m are Gaussian integers and x is complex.)

Coming back to theorem 1, a very nice solution of it can be obtained by using a technique similar to the one used in **[1]**.

Lemma 3: Suppose we have set A consisting of a elements and its subset B consisting of b elements. If we choose a subset B' from A consisting of b' elements (with elements in B' chosen at random). The expected number of elements in intersection of B and B' is equal to $b' + \frac{b}{a} = b'\frac{b}{a}$.

Theorem 3: (Dirichlet's Theorem in Complex Numbers): $\sum_{N(p) \leq N(x)}^{p \in ak+b} 1 \sim \frac{N(x)}{\phi(a)(N(\log x))}$.

The proof for this can be found in **[2]**

rhetoric, able to contend —.Francis Bacon

Section 1. Consider the following 2 pairs of arithmetic progressions:

$$((1+i)k+0),\ ((1+i)k+m)$$

$$((1+i)k+1),\ ((1+i)k+m+1)$$

where $k = 0, 1, 2, 3, \ldots$

All the primes which differ by m will fall in the second pair $((1+i)k+1)$, $((1+i)k+m+1)$, since $(1+i)k$ is divisible by $(1+i)$. Therefore, the question of how many primes differing by m exist with norm up to $N(x)$ exist up to a certain x is equivalent to finding out how many values of k make $(1+i)k+1$ and $(1+i)k+m+1$ simultaneously prime.

If we assume that any value of k is equally likely to make $(1+i)k+1$ prime, and any value of k is equally likely to make $(1+i)k+m+1$ prime (this assumption is incorrect and will be improved on shortly). Further, assume that the primes belonging to $(1+i)k+1$ behave independently from the primes belonging to $(1+i)k+m+1$. Thus, using Lemma 3 and Theorem 3 we have that

$$p_m(x) \sim \frac{\left(\frac{N(x)}{\phi(1+i)N(\log x)}\right)^2}{N\left(x/(1+i)\right)} = 2\frac{N(x)}{(N(\log x))^2}$$

In the above, $b \sim b' \sim \left(\frac{N(x)}{\phi(1+i)N(\log x)}\right)$, and $a = \left[\frac{N(x)}{N(1+i)}\right] = \frac{N(x)}{N(1+i)}$.

The $N(x)/(N(\log x))^2$ reflects the conjecture, but the constant $(1+i)$ does not. The assumption that any value of k is equally likely to make $(1+i)k+1$ prime, and that any value of k is equally likely to make $(1+i)k+m+$ is incorrect. For, if $k \equiv 1 \bmod (1+2i)$ (which is the prime with the next highest norm excluding 1 - i which is the conjugate of $(1+i)$ and which I am going to speak about later) then $(1+i)k+1$ is composite (since it is divisible by $1+2i$ when $k>1$). And so, we improve on the above result by considering pairs of arithmetic progressions $\bmod (1+i)(1+2i) = (-1+3i)$ rather than pairs of arithmetical progressions $\bmod (1+i)$.

And so we continue in the above fashion indefinitely, and find that

$$p_m(\text{x}) \sim \lim_{k\to\infty} \phi_m((1+i)(1+2i)3(2+3i)\ldots p_k)\left(\frac{\left(\frac{N(x)}{\phi((1+i)(1+2i)3(2+3i)\ldots p_k)N(\log x)}\right)^2}{\frac{N(x)}{N((1+i)(1+2i)3(2+3i)\ldots p_k)}}\right)$$

$$= 2C_i \prod_{p>2}^{p|m} \frac{p-1}{p-2} \frac{N(x)}{(N(\log x))^2} \times$$

$$\lim_{k\to\infty} \frac{\phi_m((1+i)(1+2i)3(2+3i)\ldots p_k)N((1+i)(1+2i)3(2+3i)\ldots p_k)}{(\phi((1+i)(1+2i)3(2+3i)\ldots p_k))^2} \times \frac{N(x)}{(N(\log x))^2}$$

The law embodies the story of a nation's development through many centuries, and it cannot be dealt with as if it where $f_m(n)$ denotes the number of pairs $c, c+m$ such that $(n, c) = (n, c+m) = 1$, and $0 \le N(c) \le N(n)$ (i.e. the number of pairs of arithmetical progressions $(nk+c)$, $(nk+c+m)$ that contain infinitely many primes). The problem is reduced to finding the suitable formula for $f_m((1+i)(1+2i)3((2+3i)\ \ldots\ p^k)$.

Section 2. Let's start seeking the formula by first considering $\phi_m(p^k)$ where p is prime. If $(p,m)=1$ then, of Gaussian integers 0, 1, $1+i$, 1 - i, 2, . . . l, where l is the Gaussian integers with the largest norm smaller than p^k, there are exactly $N(p^{k-1})$ integers that are not relatively prime to p^k (namely 0, p, $(1+i)p$, $(1-i)p$, $2p$, . . . , $(p^{k-1}-1)p$). Each of those p^{k-1} Gaussian integers kills two arithmetic progressions and so

$\phi_m(p^k) = N(p^k) - 2N(p^{k-1}) = N(p^k - 1)(N(p) - 2)$, if $(p,m)=1$. On the other hand, if $p|m$, then each multiple of p only kills one pair of arithmetic progressions (since we must not double count) and so $\phi_m(p^k) = N(p^k) - N(p^{k-1}) = N(p^k - 1)(N(p) - 1)$, if $p|m$.

Now that we have the formula for $\phi_m(p^k)$ where p is prime, the next step is to show that ϕ_m is multiplicative, that is, $\phi_m(a)\phi_m(b) = \phi_m(ab)$. To do that I'm going to prove the statement which is more general, $\phi_{(n,m)}(a)\phi_{(n,m)}(b) = \phi_{(n,m)}(ab$, where n is a nonnegative integer. $\phi_{(n,m)}$ (a) denotes the number of pairs c, $nc+m$ such that $(a,c) = (a, nc+m) = 1$, and $0 \le N(c) \le N(n)$ (i.e. the number of pairs of arithmetical progressions $(ak+c, n(ak+c)+m)$ that contain infinitely many primes). Clearly, when $n=0$, our function becomes simply $\phi_m(a)$, multiplicity of which establishes the relationship which we seek. Here is the proof:

Lemma 4: $\phi_{(n,m)}(a,b) = \phi_{(n,m)}(a)\phi_{(n,m)}(b)$ if $(a,b)=1$.

And that is exactly what we needed to prove for our lemma.

Using our new and old results, we can now obtain the formula for the prime density:

$$p_m(x) \sim \prod_{N(p)>2}^{p|m} \frac{N(p)-1)}{(N(p)-2)} \frac{N(x)}{(N(\log x))^2}.$$

Section 3: It seems like we have obtained the formula for pairs of primes differing by a Gaussian integer. However, there is something that was left out. We did not count some primes. As a matter of fact, we did not count any primes which were conjugates of the primes we used (i.e. 1+i, 2-i, 3-2i...) To be consistent, we'll have to consider these primes too. Well, suppose we add all the conjugates to the primes in Section 1. This is what we'll get (we don't count $1-i$ since it is a conjugate of $1+i$):

$$\phi((1+i)(1+2i)(1-2i)3(2+3i)(2-3i)\ldots p^k) = \phi_m((1+i)\times 5\times 3\times 13\times 17\ldots p^k).$$

Since all the conjugates will multiply out to give integers, we'll have the product of all primes in parenthesis (though, the order will be quite strange at times. Since $N(4+i) = N(4-i) = 17 < 49 = N(7)$, 7 will appear much later in the sequence than 17. The same is true of all primes p, such that. $p \equiv 3 \bmod 4$ since they cannot be represented as sum of two squares or product of conjugates.

contained only the axioms and corollaries of a book of mathematics. — Oliver Wendell Holmes Jr.

Thus, $p_m(x) \sim \lim\limits_{k\to\infty} \frac{\phi_m((1+i)\cdot 5\cdot 3\cdot 13 \ldots p_k) N((1+i)\cdot 5\cdot 3\cdot 13 \ldots p_k)}{(\phi((1+i)\cdot 5\cdot 3\cdot 13 \ldots p_k))^2} \frac{N(x)}{(N(\log x))^2} = 2C_i \prod_{p>2}^{p|m} \frac{p-1}{p-2} \frac{N(x)}{(N(\log x))^2}$,

where $C_i = \prod_{p>2} (p - \frac{p}{(p-1)^2})$. Clearly this is a much nicer result to deal with (not mentioning the more accurate), since except for $N(x)$ and $N(\log x)^2$ all the calculations are done with integers, thus greatly simplifying the task of finding $p_m(x)$ for some particular x. Some amazing conclusions could already be seen at this point, but I'll leave them till the end of the work on Theorem 4.

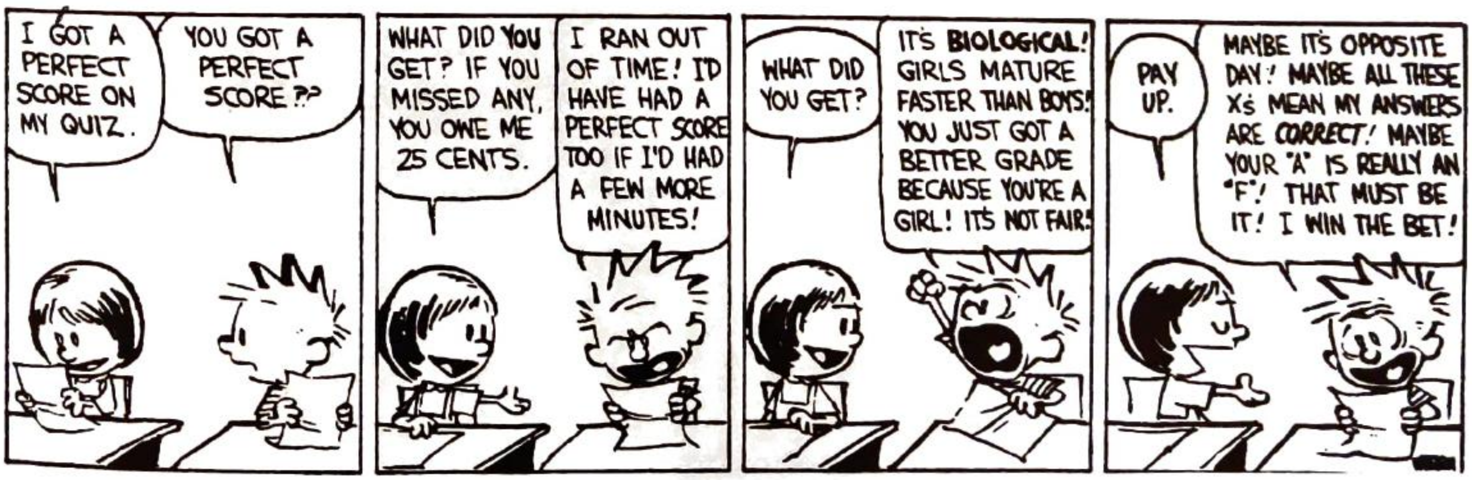
Section 4. Now, I'll attempt generalizing the obtained result for primes differing linearly. Thus, the formula will be almost identical:

Theorem 4. $p_{(m,n)}(x) \sim 2C_i \prod_{p>2}^{p|m \text{ or } p|n} (\frac{N(p)-1}{N(p)-2}) \frac{N(x)}{(N(\log x))^2}$ (m and n must be odd to avoid division by 0)

As you can see, the result looks very nice, but if looked at carefully, it can be seen that the result is something that could have been expected, especially noticing that in the plane (habitat of Gaussian integers) the function for primes on the line (habitat of integers) would have to be squared, and the relation between the formula in Theorem 4 and the similar formula for regular integers, which could be derived easily, is exactly that: quadratic!

Conclusion

Many results previously done in integers can be generalized to the appropriate systems of cyclotomic integers. For example, the proof above for Gaussian integers can be generalized to the system **Z** [Ö2] (which is the same system as **G,** except *i* is now the square root of 2) and to many other systems where UFT holds. The results such as the research above can pave the way for many similar generalizations to these systems and also can prove to be very helpful with dealing with problems in the integers (since **G** contains **Z** in itself). The reader is encouraged to do future research in this field.



If in other sciences we should arrive at certainty without doubt and truth without error, it behooves us to place the

Regents Review

Due to the New York State Board of Regents' introduction of a new high school mathematics curriculum, it is more difficult for us to provide overview of the material covered than in previous years. Readers who are in the classic courses - algebra, geometry, and trigonometry - should have no difficulty in recognizing the material required for their exams. For the benefit of those in sequential math - MQ1/2, MQ3/4, and MQ5/6 - we have prefixed the appropriate sections with roman numerals, thus indicating the year in which the material is taught.

This section does not pretend to be a complete review. It includes only the most important theorems and formula.

(II, III) Equations of Plane Curves

1. Line
   a. $y = mx + b$ where m = slope and b = y-intercept.
   b. $y - y_1 = m(x - x_1)$ where (x_1, y_1) is a point on the line and m is the slope.
   c. Slopes of parallel lines are equal.
   d. The product of the slopes of two perpendicular lines is -1.
2. Circle
   $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle with radius r.
3. Ellipse
   $x^2/a^2 + y^2/b^2 = 1$, where a and b are the major and minor axes.
4. Parabola
   a. $(x - h)^2 = 2p(y - k)$, where (h, k) is the vertex and p is the distance from the vertex to the directrix
   b. $y = ax^2 + bx + c$, where $x = -b/2a$ is the axis of symmetry.
5. Hyperbola
   $x^2/a^2 - y^2/b^2 = 1$, where a and b are the major and minor axes.

(III) Logarithms

1. $\log_b N = E$ means $b^E = N$.
2. $\log_c 1 = 0$
3. $\log_c c = 1$
4. $\log_c c^b = b$
5. $\log_c a^b = b \log_c a$
6. $\log_c ab = \log_c a + \log_c b$
7. $\log_c 1/a = -\log_c a$
8. $\log_c (a/b) = \log_c a - \log_c b$
9. $\log_c \sqrt[b]{a} = (\log_c a)/b$

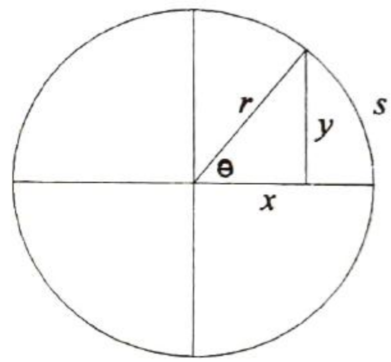
(II) Laws of Exponents

1. $x^m\ x^n = x^{(m+n)}$
2. $(x^m)^n = x^{mn}$
3. $x^m/x^n = x^{(m-n)}$
4. $x^{-m} = 1/x^m$
5. $x^0 = 1$
6. $x^{1/m} = \sqrt[m]{x}$
7. $m^{m/n} = \sqrt[n]{x^m} = \left(\sqrt[n]{x}\right)^m$

(II) The Quadratic Formula

1. The roots of the equation $ax^2 + bx + c$ are $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
2. If the discriminant, $b^2 - 4ac$, is
   a. positive, the roots are real and distinct.
   b. zero, the roots are real and equal.
   c. negative, the roots are complex conjugates.
3 Sum of roots = $-b/a$
4. Product of roots = c/a

(III) Trigonometry



1. SOH - CAH - TOA
   a. sin - opposite over hypotenuse
   b. cos - adjacent over hypotenuse
   c. tan - opposite over adjacent
2. $\sin \theta = y/r$
3. $\cos \theta = x/r$
4. $\tan \theta = y/x$
5. $\csc \theta = r/y = 1/\sin \theta$
6. $\sec \theta = r/x = 1/\cos \theta$
7. $\cot \theta = x/y = 1/\tan \theta$
8. $\sin^2 \theta + \cos^2 \theta = 1$
9. $\tan^2 \theta + 1 = \sec^2 \theta$
10. $\cot^2 \theta + 1 = \csc^2 \theta$
11. $\sin \theta = \cos(90^\circ - \theta)$
12. $\tan \theta = \cot(90^\circ - \theta)$
13. $\sec \theta = \csc(90^\circ - \theta)$
14. $s = r\theta$
15. $r^2 = x^2 + y^2$
16. Sum/Difference Formulas:
   a. $\sin(x \pm y) = \sin x \cos y \pm \sin y \cos x$
   b. $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
   c. $\tan(x \pm y) = (\tan x \pm \tan y) / (1 \mp \tan x \tan y)$
17. Double Angle Formulas:
   a. $\sin(2x) = 2 \sin x \cos x$
   b. $\cos(2x) = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$
   c. $\tan(2x) = 2 \tan x / (1 - \tan^2 x)$
   (note: the double angle formula can be derived from the sum formula by letting $x = y$.)
18. Half Angle Formulas:
   a. $\sin^2 (x / 2) = (1 - \cos x) / 2$
   b. $\cos^2 (x / 2) = (1 + \cos x) / 2$
   c. $\tan^2 (x / 2) = (1 - \cos x) / (1 + \cos x)$
19. In the graph of $y = N \sin (rx)$, the amplitude is N, and the period is $2\pi/r$.
20. $360^\circ = 2\pi$ radians.
   a. To convert from degrees to radians, multiply by $\pi/180$.
   b. To convert from the radians to degrees, multiply by $180/\pi$.

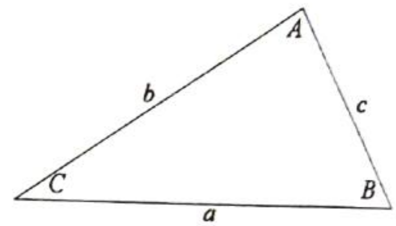
Triangles

1. Area $= \frac{1}{2}ab\sin C = \frac{1}{4}\sqrt{(a+b+c)(a+b-c)(b+c-a)(c+a-b)}$.

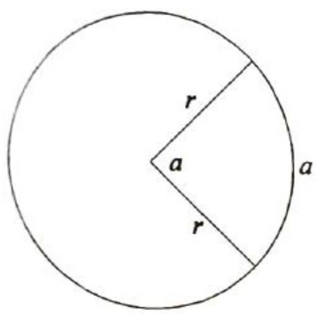
2. Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ = diameter of circumcircle

3. Law of Cosines: $c^2 = a^2 + b^2 - 2\,ab\cos C$

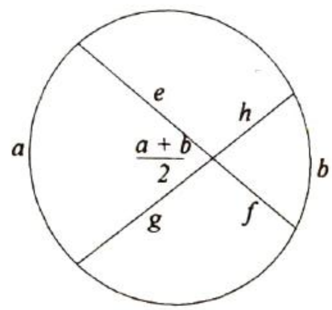
4. Law of Tangents: $\frac{\tan\frac{A+B}{2}}{\tan\frac{A-B}{2}} = \frac{a+b}{a-b}$.



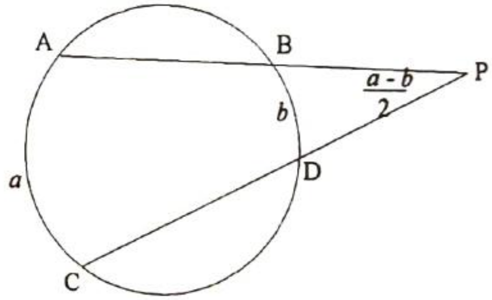
(III) Circles



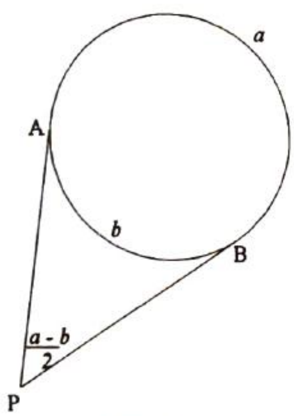
r = radius



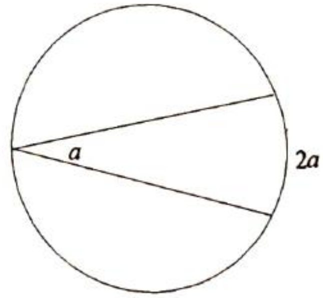
$ef = gh$

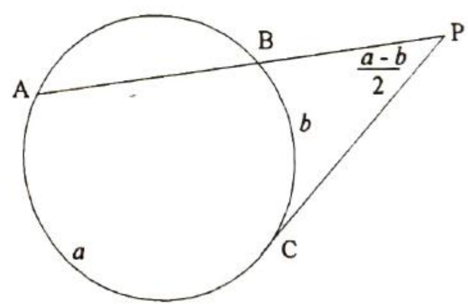


(PA)(PB) = (PC)(PD)



PA = PB





$(PA)(PB) = (PC)^2$

(II, III) Quadrilateral

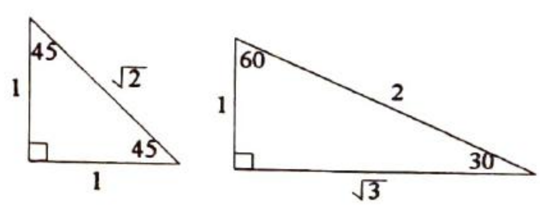
1. Two sides parallel and congruent ⇔ parallelogram
2. Both opposite sides are parallel ⇔ parallelogram
3. Diagonals bisect each other ⇔ parallelogram
4. Diagonals are perpendicular bisectors ⇔ rhombus

(I, II, III) Miscellaneous

1. Sum of interior angles of an n-gon is 180 (n - 2)
2. Each interior angle of a regular n-gon is 180 (n - 2) / n
3. Vertical angles, alternate interior angles, and base angles of isosceles triangle and trapezoid are congruent.
4. Complementary angles add up to 90°, supplementary to 180°.

   (II) (A = Angle, S = Side, b = base, h = altitude)
5. Two triangles are congruent if you can prove *SSS*, *SAA*, *SAS*, or *ASA*.
6. Two triangles are similar if you can prove *AA*.
7. Two right triangles are congruent if you can prove their hypotenuses are congruent and one pair of legs are congruent. (*HL*)
8. The sum of the angles in a triangle is 180°.
9. The base angles of an isosceles triangle are congruent.
10. Area of triangle = (1/2) bh.
11. Area of a trapezoid = (1/2) $(b_1 + b_2)\,h$.
12. Area of a parallelogram = bh.
13. Area of a regular n-gon = (1/2) ap (where a = apothem, p = perimeter).
14. In a right triangle with legs a and b and hypotenuse c, $a^2 + b^2 = c^2$.
15. In a circle, the diameter that is perpendicular to a chord bisects the chord.
16. The locus of points at distance r from point P is the circle with center P and radius r.
17. The locus of points equidistant from two point P and Q is the perpendicular bisector of PQ.
18. The locus of points equidistant from two lines l and m are the bisectors of the angles formed by l and m.

   (B = area of base, h = height, r = radius)
19. Volume of a prism = Bh.
20. Volume of a cylinder = $\pi r^2 h$.
21. Volume of a pyramid = (1/3) Bh.
22. Volume of a cone = (1/3) $\pi r^2 h$
22. Volume of a sphere = (4/3) πr^3



(I, III) Probability:

1. Let A and B represent events in a sample space.
2. If A and B are independent, then $P(A \cap B) = P(A) \times P(B)$.
3. If A and B are disjoint, then $P(A \cap B) = 0$ and $P(A \cup B) = P(A) + P(B)$.
4. If the probabilities are uniformly distributed, event A can occur in $n(A)$ ways, and there are $n(S)$ possible outcomes in S, then $P(A) = n(A)/n(S)$.
5. The number of permutations of n objects is $n! = n\,(n - 1)\ldots(2)\,(1)$.
6. The number of permutations of n objects, k of which are indistinguishable, is $n!\,/\,k!$.
7. The number of permutations of r objects chosen from a set of n objects is ${}_nP_r = \dfrac{n!}{(n-r)!}$

   $= n\,(n - 1)\,(n - 2)\ldots(n - r + 1)$.
8. The number of combinations of r objects chosen from a set of n objects is ${}_nC_r = \dbinom{n}{r} = \dfrac{n!}{r!\,(n-r)!} = {}_nP_r$

   $/\,r!$.

(III) The binomial theorem

$$(a+b)^n = \sum_{r=0}^{n}\binom{n}{r}a^r b^{n-r} = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b + \cdots + \binom{n}{n-1}a^1 b^{n-1} + b^n$$

(I, II) Logic.

Laws of Inference	Symbolic Forms
1. Law of Detachment (Modus Ponens)	$[(p \Rightarrow q) \wedge p] \Rightarrow q$
2. Law of the Contrapositive	$(p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$.
3. Law of Modus Tollens	$[(p \Rightarrow q) \wedge \sim q] \Rightarrow \sim p$
4. Chain Rule (Law of Syllogism)	$[(p \Rightarrow q) \wedge (q \Rightarrow \mathrm{r})] \Rightarrow (p \Rightarrow r)$
5. Law of Disjunctive Inference	$[(p \vee q) \wedge \sim p] \Rightarrow q$ $[(p \vee q) \wedge \sim q] \Rightarrow p$
6. Law of Double Negation	$\sim(\sim p) \Leftrightarrow p$
7. De Morgan's Laws	$\sim(p \wedge q) \Leftrightarrow (\sim p \vee \sim q)$ $\sim(p \vee q) \Leftrightarrow (\sim p \wedge \sim q)$
8. Law of Simplification	$(p \wedge q) \Rightarrow p$ $(p \wedge q) \Rightarrow q$
9. Law of Conjunction	$(p) \wedge (q) \Rightarrow (p \wedge q)$
10. Law of Disjunctive Addition	$p \Rightarrow (p \vee q)$

(I, III) Statistics

1. Measures of Central Tendency
   a. The mean, or arithmetic mean, of a set of n numbers is the sum of the numbers divided by n

      Therefore, for a sample of scores, $x_1, x_2, x_3, \ldots, x_n$, the mean $\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$.
   b. When a set of data is arranged in numerical order, the "middle" score is called the median. If there is an even number of terms, then the medians is found by averaging the two "middle" terms.
   c. The mode is the score that appears most often in a set of data. For some sets of data there can be more than one mode (when two modes appear, the data is called bimodal) and, in some cases, no mode whatsoever.
2. Measures of Dispersion
   a. The range of a set of data is the difference between the highest and the lowest scores.
   b. The mean absolute deviation of a set of data is the average value of the difference between the data

      and the mean. Mean deviation $= \frac{1}{n}\sum_{i=1}^{n}\left|x_i - \bar{x}\right|$.
   c. The variance, v, of a set of data is the average of the squares of the deviations from the mean.

      Variance $v = \frac{1}{n}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2$
   d. The standard deviation, s, is a set of data is equal to the square root of the variance. Standard

      deviation $s = \sqrt{v} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2}$

(III) Functions

1. $y = f(x)$: x is domain, y is range.
2. $f \circ g(x) = f(g(x))$, x is domain of g, g is domain of f.
3. f^{1} of $f(x)$: $f \circ f^{1}(x) = x$.
4. Symmetry
   a. If $f(x) = f(-x)$ then f is symmetric about y axis.
   b. If $f(x) = -f(-x)$ then f is symmetric about origin.
5. f is a function if and only if no vertical line passes through more than one point of the graph of f.

(I, II, III) Transformations

1. Line reflections
   a. respect to x-axis: $(x, y) \Rightarrow (-x, y)$
   b. respect to y-axis: $(x, y) \Rightarrow (x, -y)$
   c. respect to $y = x$: $(x, y) \Rightarrow (y, x)$
2. Point reflections: origin: $(x, y) \Rightarrow (-x, -y)$
3. Translations: $T_{(h,k)}\,(x, y) \Rightarrow (x + h, y + k)$
4. Rotations
   a. 90° counterclockwise: $(x, y) \Rightarrow (-y, x)$
   b. 90° clockwise: $(x, y) \Rightarrow (y, -x)$
5. Dilations: $D_k\,(x,y) \Rightarrow (kx, ky)$

A.P. Calculus Review

Originally by David Z. Weinstein, Updated by *Math Survey* Staff

All material is covered by BC calculus; asterisks (*) indicate material that does not apply to AB calculus.

Limits

1.* Defn: Let $f(x)$ be defined in some open interval containing a (except possibly at a itself), then $\lim_{x\to a} f(x) = L$, if for any $\varepsilon > 0$ (however small) there exist a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ ($L - \varepsilon < f(x) < L + \varepsilon$) whenever $0 < |x - a| < \delta$ ($a - \delta < x < a + \delta$).

   a.* Defn: The statement $\lim_{x\to\infty} f(x) = L$ means that for every $\varepsilon > 0$, there exists an $M > 0$ such that $|f(x) - L| < \varepsilon$ whenever $x > M$.

   b.* Defn: The statement $\lim_{x\to-\infty} f(x) = L$ means that for every $\varepsilon > 0$, there exists an $N < 0$ such that $|f(x) - L| < \varepsilon$ whenever $x < N$.

   c.* Defn: The statement $\lim_{x\to c} f(x) = \infty$ means that for each $M > 0$, there exists a $\delta > 0$ such that $f(x) > M$ whenever $0 < |x - c| < \delta$.

   d.* Defn: The statement $\lim_{x\to c} f(x) = -\infty$ means that for each $N < 0$, there exists a $\delta > 0$ such that $f(x) < N$ whenever $0 < |x - c| < \delta$.

2. Limit May Fail to Exist if

   a. $\lim_{x\to a^+} f(x) \neq \lim_{x\to a^-} f(x)$ (e.g. $\lim_{x\to 0}\frac{|x|}{x}$).

   b. values of $f(x) \to \infty$ (e.g. $\lim_{x\to 0}\frac{1}{x}$).

   c. values of $f(x)$ may oscillate indefinitely, approaching no limit (e.g. $\lim_{x\to 0}\sin\frac{1}{x}$).

3. Basic Limit Theorems:
   a. $\lim\,[f(x) \pm g(x)] = \lim f(x) \pm \lim g(x)$.
   b. $\lim\,[f(x)\cdot g(x)] = \lim f(x)\cdot\lim g(x)$.
   c. $\lim\,[f(x)/g(x)] = \lim f(x)/\lim g(x)$ ($\lim g(x) \neq$ to 0).
   d. $\lim\,[\sqrt[n]{f(x)}\,] = \sqrt[n]{\lim f(x)}$.

Continuity

1. Defn: f is continuous at a provided $f(a)$ is defined, $\lim_{x\to a} f(x)$ exists, and $\lim_{x\to a} f(x) = f(a)$ (equivalently: $f(x)$ is continuous at $x = a$, if for any $\varepsilon > 0$, there exists a $\delta > 0$, such that $|f(x) - f(a)| < \varepsilon$ whenever $|x - a| < \delta$.
2. Important Limit Theorems
   a. If f and g are continuous functions, then fg, cf, $f \pm g$, f/g ($g \neq 0$) are also continuous.
   b. All polynomials are continuous.

c. A function continuous on a closed interval $[a, b]$ has the following properties.
   i. $f(x)$ has a minimum and maximum value on the interval.
   ii. $f(x)$ cannot pass from one value to another in the interval without taking every intermediate value (i.e. It's an unbroken line).

Derivatives

1. Def'n: let $x = a$ be on the curve $y = f(x)$. At each such point where $\lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$ exists, the function is said to be differentiable (have a derivative). The limit is called the derivative of y with respect to x and denoted by $\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$. Additionally, $f(a) = \lim_{\Delta x \to 0} \frac{f(a+\Delta x) - f(a)}{\Delta x} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ (derivative at a). If $y = f(x)$ then
   a. $dy/dx = f'(x) = D_x y = y'$ (first derivative).
   b. $d^2y/dx^2 = f''(x) = D^2_x\, y = y''$ (second derivative).
   c. $d^3y/dx^3 = f'''(x) = D^3_x\, y = y'''$ (third derivative).
2. Formulas for Derivatives
   c, n are constants; u, v are differentiable functions of x.

1. $\frac{d(c)}{dx} = 0$

2. $\frac{d(cu)}{dx} = c\frac{du}{dx}$

3. $\frac{d(u^n)}{dx} = nu^{n-1}\frac{du}{dx}$

4. $\frac{d(u+v)}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

5. $\frac{d(uv)}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

6. $\frac{d\left(\frac{u}{v}\right)}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

7. $\frac{d(\sin u)}{dx} = \cos u\frac{du}{dx}$

8. $\frac{d(\cos u)}{dx} = -\sin u\frac{du}{dx}$

9. $\frac{d(\tan u)}{dx} = \sec^2 u\frac{du}{dx}$

10. $\frac{d(\cot u)}{dx} = -\csc^2 u\frac{du}{dx}$

11. $\frac{d(\sec u)}{dx} = \sec u \tan u\frac{du}{dx}$

12. $\frac{d(\csc u)}{dx} = -\csc u \cot u\frac{du}{dx}$

13. $\frac{d(\ln u)}{dx} = \frac{1}{u}\frac{du}{dx}$

14. $\frac{d(e^u)}{dx} = e^u\frac{du}{dx}$

15. $\frac{d(\log_c u)}{dx} = \log_c e \cdot \frac{1}{u}\frac{du}{dx}$

16. $\frac{d(a^u)}{dx} = a^u \ln a\frac{du}{dx}$

17. $\frac{d(\sin^{-1} u)}{dx} = \frac{1}{\sqrt{1-u^2}}\frac{du}{dx}$

18. $\frac{d(\cos^{-1} u)}{dx} = -\frac{1}{\sqrt{1-u^2}}\frac{du}{dx}$

19. $\frac{d(\tan^{-1} u)}{dx} = \frac{1}{1+u^2}\frac{du}{dx}$

20. $\frac{d(\cot^{-1} u)}{dx} = -\frac{1}{1+u^2}\frac{du}{dx}$

21. $\frac{d(\sec^{-1} u)}{dx} = \frac{1}{\sqrt{u^2(u^2-1)}}\frac{du}{dx}$

22. $\frac{d(\csc^{-1} u)}{dx} = -\frac{1}{\sqrt{u^2(u^2-1)}}\frac{du}{dx}$

23. **Chain Rule (for composite functions)**

If $y = f(u)$ and $u = g(x)$ $(y = f(g(x)))$, then $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = f'(u)g'(x) = f'(g(x))g'(x)$.

24. If $y = f(x)$ is single-valued and has nonzero derivatives on a given interval, then the inverse function exists and has a derivative. Suppose $y = f(x)$ has an inverse function (i.e. $x = f^{-1}(y)$ exists) that is differentiable, then $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$.

25.***Parametric Equations**
If $y = f(t)$ and $x = g(t)$, then

a.* $y' = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$.

b.* $y'' = \frac{d^2y}{dx^2} = \frac{dy'}{dx} = \frac{\frac{dy'}{dt}}{\frac{dx}{dt}}$.

c.* $y''' = \frac{d^3y}{dx^3} = \frac{dy''}{dx} = \frac{\frac{dy''}{dt}}{\frac{dx}{dt}}$.

26. **Logarithmic Differentiation**

If $y = x^{g(x)}$ then $\ln y = \ln x^{g(x)} = g(x) \ln x$. Differentiating yields $\frac{1}{y}\frac{dy}{dx} = g'(x)\ln x + g(x)\frac{1}{x}$ or

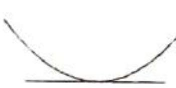
$$\frac{dy}{dx} = y\left(g'(x)\ln x + g(x)\frac{1}{x}\right).$$

Curve Sketching

1. Without Derivatives
   a. Intercepts: To get the x intercepts, set $y = 0$ and solve for x; to get the y intercepts, set $x = 0$ and solve for y.
   b. Symmetry: If a curve has form $y = F(x)$ or $f(x, y) = 0$, then it is
      i. Symmetric to x-axis if $f(x, y) = f(x, -y)$ (or $F(x) = -F(x)$).
      ii. Symmetric to y-axis if $f(x, y) = f(-x, y)$ (or $F(x) = F(-x)$).
      iii. Symmetric to origin if $f(x, y) = f(-x, -y)$ (or $F(x) = -F(x)$).
   c. Extent (domain and range): Find x and y values on which the function is defined.
   d. Asymptotes:
      i. Horizontal Asymptote: Let $x \to \pm\infty$ and find the limiting value(s) of y.
      ii. Vertical Asymptote: Let $y \to \pm\infty$ and find the limiting value(s) of x.
      iii. Oblique Asymptote $y = mx + b$ $(m \neq 0)$ is an asymptote of $y = f(x)$ if $\lim_{x\to\infty}[f(x) - (mx + b)] = 0$.
      iv. We can also solve for one variable in terms of the other; set denominator equal to 0, etc. For

example, if $y = f(x) = g(x) + \dfrac{c}{h(x)}$, then $y = g(x)$ is a curved (e.g. parabolic, oblique, or horizontal) asymptote, and $h(x) = 0$ represents any vertical asymptotes.

2. With Derivatives
   a. Def'n:
      i. A function $f(x)$ is increasing (decreasing) on an interval if whenever x_1 and x_2 belong to the interval and $x_2 > x_1$, then either $f(x_2) > f(x_1)$ (rising curve) or $f(x_2) < f(x_1)$ (falling curve).
      ii. $f(x)$ is a smooth function if both $f(x)$ and $f'(x)$ are continuous.
      iii. Relative Extremum: maximum or minimum value.
      iv. Critical values of $f(x)$ are values where $f'(x) = 0$ or is undefined.
      v. A smooth curve is concave upward at a point if successive tangents to the curve from that point turn counterclockwise (slope increases - curve lies above tangent). A smooth curve is concave downward at a point if successive tangents to the curve from that point turn clockwise (slope decreases - curve lies below tangent).
      vi. Point of inflection: a point where the curve changes its concavity.



   b. Important Theorems:
      i. Function $f(x)$ is increasing at $x = a$ if $f'(a) > 0$, decreasing at $x = a$ if $f'(a) < 0$.
      ii. If $f(x)$, a continuous, differentiable function, has a relative extrema at $x = a$, then $f'(a) = 0$ (but not conversely).
      iii. Tests for Maxima and Minima
         (1) Max: $f'(a) = 0$ and $f'(x)$ changes from + to - at $x = a$.
         Min: $f'(a) = 0$ and $f'(x)$ changes from - to + at $x = a$.
         (2) Max: $f'(a) = 0$ and $f''(a) < 0$.
         Min: $f'(a) = 0$ and $f''(a) > 0$.

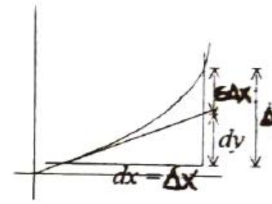
      iv. Curve is concave up at $x = a$ if $f''(a) > 0$; concave down if $f''(a) < 0$. (If $f''(a) = 0$, it's concave up or down depending on its concavity in the neighborhood of $x = a$.)

      v. If $f(x)$ is a function with continuous second derivative, $x = a$ is a point of inflection of $f(x)$, if $f''(a) = 0$ and $f(a) \neq 0$ or $f''(x)$ changes sign through a.
   c. Absolute extrema can be attained
      i. Where $f'(x) = 0$.
      ii. At the endpoints of interval.
      iii. Where the derivative does not exist.

Applications of the Derivative

1. If a function is differentiable at a point, then it must be continous at the point, but not conversely, i.e.: differentiability implies continuity.
2. Rolle's theorem: If $f(x)$ is continuous on $[a, b]$, $f'(x)$ exists on (a, b), and $f(a) = (b) = 0$, then there exists a point c $(a < c < b)$ such that $f'(c) = 0$.
3. Mean value theorem for derivatives: If $f(x)$ is continuous on $[a, b]$ and $f'(x)$ exists on (a, b) then there exists a point c such that $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.

4. **L'Hôpital's Rule**
   a. If $f(x)$ and $g(x)$ are in an interval including $x = a$ and satisfy the hypotheses of the general mean value theorem and $\lim_{x\to a}\dfrac{f(x)}{g(x)} = \dfrac{0}{0}$ or $\dfrac{\infty}{\infty}$, then $\lim_{x\to a}\dfrac{f(x)}{g(x)} = \lim_{x\to a}\dfrac{f'(x)}{g'(x)}$.
   b. Evaluation of other indeterminate forms:
      i. $0 \cdot \infty$. Change to $(1/\infty)(\infty)$, which equals ∞/∞.
      ii. $\infty - \infty$. Transform by algebraic or trigonometric manipulation to acceptable form.
      iii.* 1^∞, 0^0, and ∞^0. Take the log of the function and change to one of the above forms.
5. Differentails. Def'n: When $y = f(x)$, the differential of x (independent value) $dx = \Delta x$ and the differential of y (dependent value) $dy = f'(x)\,dx$.
   Motivation for definition:
   a. One may now think of $f'(x)$ as a quotient.
   b. For small Δx, $dy = \Delta y$.



Note: $f'(x) = \lim_{\Delta x\to 0}\dfrac{\Delta y}{\Delta x}$, therefore $\dfrac{\Delta y}{\Delta x} = f'(x) + \varepsilon$ ($\varepsilon \to 0$ when $\Delta x \to 0$) implies $\Delta y = f'(x)\Delta x + \varepsilon\Delta x$. Main part of Δy is $f'(x)\Delta x$ since $\varepsilon\Delta x$ is relatively small. So we define $dy = f'(x)\Delta x$, therefore $\Delta y = dy + \varepsilon\Delta x \approx dy$.

6. Related rates problems: Find the relationship between variables, take the derivative with respect to time, and substitute.
7. Newton's Method for approximating zeros of functions: Let $f(c) = 0$, where f is differentiable on an open interval containing c. Then, to approximate c, we use the following steps:
   a. Make an initial estimate x_1, that is close to c. (A graph is helpful.)
   b. Determine a new approximation by $x_{n+1} = x_n - f(x_n) / f'(x_n)$.
   c. If $|x_n - x_{n+1}|$ is less than the desired accuracy, let x_{n+1} serve as the final approximation. Otherwise, return to step b and calculate a new approximation.

Motion of Particles

1. Rectilinear motion (motion along a straight line)
   If $s = f(t)$, then $v_{Av} = \Delta s/\Delta t$ = (total distance traveled) / (time elapsed) and $a_{Av} = \Delta v/\Delta t$.
   a. $v = ds/dt = f'(t)$. If $v > 0$, then the particle is moving in the direction of increasing s. If $v < 0$, then the particle is moving in direction of decreasing s. If $v = 0$, then the particle is instanteously at rest. At the instant the particle reverses direction, v must be 0, but not conversely.
   b. $a = dv/dt = f''(t)$. The particle changes direction if $v = 0$ and $a \neq 0$ or v changes sign as we go through $v = 0$. The speed of the particle, $|v|$ (magnitude of v), increases when $a > 0$ and $v > 0$ or $a < 0$ and $v < 0$. The speed decreases when $a < 0$ and $v > 0$ or $a > 0$ and $v < 0$.
2.* Curvilinear motion (motion along a curve)

   Parametric equations of motion: $x = f(t)$, $y = g(t)$; $v_x = \dfrac{dx}{dt}$, $v_y = \dfrac{dy}{dt}$, $a_x = \dfrac{dv_x}{dt}$, $a_y = \dfrac{dv_y}{dt}$.

   $\|\mathbf{v}\| = \sqrt{v_x^{\,2} + v_y^{\,2}}$ (magnitude), $\tan\theta = \dfrac{v_y}{v_x}$ (quadrant depends on signs of v_x and v_y).

   $\|\mathbf{a}\| = \sqrt{a_x^{\,2} + a_y^{\,2}}$ (magnitude), $\tan\alpha = \dfrac{a_y}{a_x}$ (quadrant depends on signs of a_x and a_y).

Since $\tan\theta = \frac{\frac{dx}{dt}}{\frac{dy}{dt}} = \frac{dy}{dx}$ and $v^2 = v_x^{\,2} + v_y^{\,2} = \left(\frac{ds}{dt}\right)^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$, $\|\mathbf{v}\| = \left|\frac{ds}{dt}\right|$.

It follows that the velocity vector has is tangent to the curve and the speed of the particle = $\left|\frac{ds}{dt}\right|$, where s is the distance along the curve from some fixed point.

Integration

1. Def'n: Indefinite Integral (Antiderivative) $\int f(x)\,dx = F(x) + C$ if $F'(x) = f(x)$.
   Fundamental Rules:
   a. $\int (u + v)\,dx = \int u\,dx + \int v\,dx$.
   b. $\int ku\,dx = k\int u\,dx$.
   c. $\int u^n\,du = u^{n+1}/(n+1) + C \quad (n \neq -1)$.
2. Def'n: Definite Integral
   Properties:

   a. $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$.

   b. $\int_a^a f(x)\,dx = 0$.

   c. $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$ (for any c).

   d. $\int_a^b k\,dx = k(b - a)$.

   e. $\int_a^b (u + v)\,dx = \int_a^b u\,dx + \int_a^b v\,dx$.
3. Fundamental Theorem of Integral Calculus (Connects concept of indefinite and definite integrals)
   If $f(x)$ is continuous on (a, b), $\int f(x)\,dx = F(x)$ for any particular integral, then

   $\int_a^b f(x)\,dx = F(b) - F(a)$.

Integration Theorems

1. If $f(x)$ is continuous on $[a, b]$, then it is integrable on $[a, b]$. (This works even for finite numbers of discontinuities.)

2. If f and g are integrable on $[a, b]$ and $f(x) \leq g(x)$ for all x on $[a, b]$, then $\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx$.

3. If f is integrable on $[a, b]$ and $m \leq f(x) \leq M$ for all x in $[a, b]$, then $m(b - a) \leq \int_a^b f(x)\,dx \leq M(b - a)$.
4. If $F(x)$ and $G(x)$ are antiderivatives of the same function (i.e. $dF/dx = dG/dx$), then $F(x) = G(x) + C$.

5. Average Value of a Function (with respect to x on $[a, b]$): $y_{av} = \frac{1}{b-a}\int_a^b f(x)\,dx$.

6. Let f be continuous on $[a, b]$. Let x be a number on (a, b) and define $F(x) = \int_a^x f(t)\, dt$, then F is differentiable and $F'(x) = f(x)$.

7. Change on Variable in Define Integrals: $x = \varphi(t)$, $a = \varphi(t_1)$, $b = \varphi(t_2)$, then $\int_a^b f(x)\, dx =$

$\int_{t_1}^{t_2} f(\varphi(t))\cdot\varphi'(t)\, dt.$

Approximations of Integrals

1. Trapezoidal Rule: $\int_a^b y\, dx \approx \frac{h}{2}(y_0 + 2y_1 + 2y_2 + \cdots + 2y_{n-1} + y_n)$, $h = \frac{b-a}{n}$.

$$\begin{array}{ll} x_0 = a & y_0 = f(x_0) = f(a) \\ x_1 = a + h & y_1 = f(x_1) \\ x_2 = a + 2h & y_2 = f(x_2) \\ \vdots & \vdots \\ x_n = a + nh & y_n = f(x_n) = f(b) \end{array}$$

$$\varepsilon_T = \frac{-(b-a)h^2 f''(c)}{\cdot 12},\ a \le c \le b.$$

2.* Area under parabolic arc: $A = (h/3)(y_0 + 4y_1 + y_2)$, where y_0, y_1, and y_2 are three equidistant ordinates, and $h = y_1 - y_0 = y_2 - y_1$.

3.* Simpson's Rule: $y = f(x) \ge 0$ on $[a, b]$.

$$\int_a^b y\, dx \approx \frac{h}{3}(y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + \cdots + 4y_{n-1} + y_n),\ h = \frac{b-a}{n},\ n \text{ must be even.}$$

$$\varepsilon_s = \frac{-(b-a)h^4 f^{(4)}(c)}{180},\ a \le c \le b..$$

Basic Integration Formulas

1. $\int c\, f(x)\, dx = c\int f(x)\, dx$.
2. $\int [f(x) + g(x)]\, dx = \int f(x)\, dx + \int g(x)\, dx$.
3. $\int u^n\, du = u^{n+1}/(n+1) + C\ \ (n \ne 1)$.
4. $\int (1/u)\, du = \ln|u| + C$.
5. $\int e^u\, du = e^u + C$.
6. $\int a^u\, du = a^u/\ln a + C\ \ (a > 0, a \ne 1)$.
7. $\int \cos u\, du = \sin u + C$.
8. $\int \sin u\, du = -\cos u + C$.
9. $\int \tan u\, du = \ln|\sec u| + C = -\ln|\cos u| + C$.
10. $\int \cot u\, du = \ln|\sin u| + C = -\ln|\csc u| + C$.
11. $\int \sec^2 u\, du = \tan u + C$.
12. $\int \csc^2 u\, du = -\cot u + C$.
13. $\int \sec u \tan u\, du = \sec u + C$.
14. $\int \csc u \cot u\, du = -\csc u + C$.
15. $\int \sec u\, du = \ln|\sec u + \tan u| + C$.
16. $\int \csc u\, du = \ln|\csc u - \cot u| + C$.
17. $\int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin\frac{u}{a} + C$.
18. $\int \frac{du}{a^2 + u^2} = \frac{1}{a}\arctan\frac{u}{a} + C$.
19. $\int \frac{du}{|u|\sqrt{a^2 - u^2}} = \frac{1}{a}\text{arcsec}\frac{u}{a} + C$.

Techniques of Integration

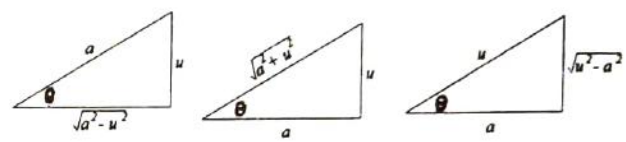
1. Substitution: Use of identities and change of variable.
2. Integration by Parts: $\int u\,dv = uv - \int v\,du$.(From integrating the product rule: $\int d(uv) = \int u\,dv + \int v\,du$.)

   Also Note: For definite integrals: $\int_a^b u\,dv = uv\Big]_a^b - \int_a^b v\,du$. Can be used for: $\int x\,e^x\,dx$, twice for $\int x^2 \cos x\,dx$, twice for $\int e^x \sin x\,dx$, $\int \ln x\,dx$, $\int \arcsin x\,dx$.
3.* Trigonometric Substitution:

   $u = a\sin\theta$ in $\sqrt{a^2-u^2}$ yields $a\cos\theta$.

   $u = a\tan\theta$ in $\sqrt{a^2+u^2}$ yields $a\sec\theta$.

   $u = a\sec\theta$ in $\sqrt{u^2-a^2}$ yields $a\tan\theta$.



4.* Partial Fractions: If the original denominator has

   a.* A linear factor, place $ax+b$ in the new denominator, with a constant numerator: $\dfrac{A}{ax+b}$.

   b.* A quadratic factor, place ax^2+bx+c in the new denominator, with a linear numerator: $\dfrac{Ax+B}{ax^2+bx+c}$.

   c.* An unfactorable polynomial to the r power, place:

$$\frac{A_{n-1}x^{n-1}+A_{n-2}x^{n-2}+\cdots+A_0x^0}{\left(a_nx^n+a_{n-1}x^{n-1}+\cdots+a_0x^0\right)^r}+\frac{B_{n-1}x^{n-1}+B_{n-2}x^{n-2}+\cdots+B_0x^0}{\left(a_nx^n+a_{n-1}x^{n-1}+\cdots+a_0x^0\right)^{r-1}}+\cdots$$

   (until the denominator is raised to the first power).

   d.* Any combinations/repetitions of the above cases should be repeated.

5. For $\int \frac{p(x)}{q(x)}$, divide the numerator by the denominator if the former is of higher or of equal degree to the latter.
6. Completing the Square: Use this technique for quadratic functions if necessary.
7. Splitting up Fractions: $\int \dfrac{Ax+B}{ax^2+bx+c} = \int \dfrac{Ax}{ax^2+bx+c} + \int \dfrac{B}{ax^2+bx+c}$.

Applications Of The Definite Integral

1. Areas:

   a. Area bounded by $y = f(x)$, x-axis, lines $x = a$ and $x = b$ $(b > a)$, $A = \int_a^b f(x)\,dx = \int_a^b y\,dx$.

   b. Area bounded by $x = g(y)$, y-axis, lines $y = a$ and $y = b$ $(b > a)$, $A = \int_a^b g(y)\,dy = \int_a^b x\,dy$.

   c. Using area of a circle $= \pi r^2$, we find $\int_0^a \sqrt{a^2-x^2}\,dx = \frac{1}{4}\pi a^2$.

   d. If $f(x) \ge g(x)$ on $[a, b]$, then the area bounded by the curves $y_1 = f(x)$, $y_2 = g(x)$ and the lines $x = a$

and $x = b$, $A = \int_a^b (y_2 - y_1)\,dx = \int_a^b (g(x) - f(x))\,dx$.

e.* Polar Coordinates:

i.* The area bounded by $r = f(\theta)$ and two radii vectors $\theta = \alpha$ and $\theta = \beta$ is $A = \frac{1}{2}\int_\alpha^\beta r^2\,d\theta$.

ii.* The area between $r_1 = f(\theta)$ and $r_2 = g(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is $A = \frac{1}{2}\int_\alpha^\beta (r_1^2 - r_2^2)\,d\theta$.

2. Volumes:

a. Known Cross-Sectional Area: Volume of solid with cross-sectional area $A(x)$ perpendicular to x-axis at x is: $V = \lim_{n\to\infty}\sum_{k=1}^{n} A(c_k)\Delta x_k = \int_a^b A(x)\,dx$.

b. Solid of Revolution: A solid generated by revolving a planar region about an axis in its plane (plane perpendicular to axis cuts circular cross sections).

c. Volume of solid formed by revolving region bounded by $y = f(x)$, $x = a$, $x = b$ $(b > a)$, about x-axis is $V = \pi\int_a^b y^2\,dx$ (Cylindrical Disks).

d. Volume of solid formed by revolving region bounded by $x = g(y)$, $y = a$, $y = b$ $(b > a)$, about y-axis is $V = \pi\int_a^b x^2\,dy$.

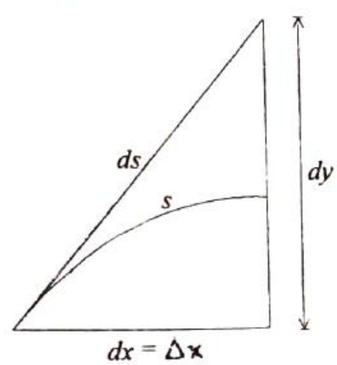
e. If $f(x) \geq g(x)$ on $[a, b]$, volume generated by revolving about x-axis region bounded by curves $y_1 = f(x)$, $y_2 = g(x)$, $x = a$ and $x = b$ $(b > a)$ is $V = \pi\int_a^b (y_2^2 - y_1^2)\,dx$ (cylindrical washers).

f. Volume of solid generated by revolving region bounded by $y = f(x)$, x-axis, $x = a$, $x = b$ $(b > a)$ about y-axis is $V = 2\pi\int_a^b xy\,dx$ (cylindrical shells). Volume of solid generated by revolving region bounded by $x = g(y)$, y-axis, $y = a$, $y = b$ about x-axis $V = 2\pi\int_a^b yx\,dy$.

3.* Arc Length:

a.* If $y = f(x)$, and $y' = f'(x)$ are continuous, then the arc length of $y = f(x)$ from $x = a$ to $x = b$ $(b > a)$ is $s = \int_a^b \sqrt{1 + y'^2}\,dx$. Similarly for $x = g(y)$, $x' = g'(y)$ from $y = a$ to $y = b$, $s = \int_a^b \sqrt{1 + x'^2}\,dy$.

b.* The differential of arc for a smooth curve $y = f(x)$, $ds = \sqrt{1 + y'^2}\,dx$. Squaring, we get $(ds)^2 = (dx)^2 + (dy)^2$ and the above formula can be written as $s = \int ds$. Also note the geometric interpretation.



c.* Parametrically, if $x = f(t)$, $y = g(t)$, then since $(ds)^2 = (dx)^2 + (dy)^2$, taking the square root and integrating yields $s = \int_{t_0}^{t_1}\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$.

d.* Polar Coordinates: Length of curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is $s = \int_{\alpha}^{\beta} \sqrt{r^2 + r'^2}\, d\theta$.

4.* Work: Motion Along Straight Line

a.* $W = fd$ (for a constant force only).

b.* Total work done in moving an object along x-axis from a to b, if $f(x)$ is a force exerted at a point x, is $W = \int_a^b f(x)\, dx$.

c.* Hooke's Law: $f \propto x$ or $f = kx$. (k is the spring constant, i.e. the force required to stretch the spring one unit from equilibrium position.)

5.* Improper Integrals

a.* Convergent: approaches a limiting value. Divergent: Approaches no value (∞, oscillates, etc.).

b.* Type 1 (Discontinuous Integrand)

i.* Integrand becomes ∞ at lower limit: $\int_a^b f(x)\, dx = \lim_{h \to 0^+} \int_{a+h}^b f(x)\, dx$.

ii.* Integrand becomes ∞ at upper limit: $\int_a^b f(x)\, dx = \lim_{h \to 0^+} \int_a^{b-h} f(x)\, dx$.

iii.* c $(a < c < b)$ makes integrand ∞: $\int_a^b f(x)\, dx = \int_a^c f(x)\, dx + \int_c^b f(x)\, dx$, use i and ii.

c.* Type 2 (Infinite Limits)

i.* $\int_a^{\infty} f(x)\, dx = \lim_{b \to \infty} \int_a^b f(x)\, dx$.

ii.* $\int_{-\infty}^b f(x)\mathrm{d}x = \lim_{a \to -\infty} \int_a^b f(x)\, dx$.

iii.* $\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{c} f(x)dx + \int_c^{\infty} f(x)\, dx$ (for suitable c).

More on Limits

1. Squeeze Theorem: If $A \le f(x) \le g(x)$ for some constant A and $\lim_{x \to c} g(x) = A$, then $\lim_{x \to c} f(x) = A$.
2. Important Limits:

a. $\lim_{x \to 0} \frac{\sin x}{x} = 1$.

b. $\lim_{x \to 0} \frac{\tan x}{x} = 1$.

c. $\lim_{x \to 0} \frac{1 - \cos x}{x} = 0$.

d. $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$.

e. $\lim_{h \to \infty} \left(1 + \frac{1}{h}\right)^h = \lim_{h \to 0} (1 + h)^{\frac{1}{h}} = e$.

f. $\lim_{k \to \infty} \left(1 + \frac{n}{k}\right)^k = \lim_{\substack{k \to \infty \\ \left(\frac{k}{n} \to \infty\right)}} \left[\left(1 + \frac{1}{\frac{k}{n}}\right)^{\frac{k}{n}}\right]^n = e^n$.

* Note: The rest of the material applies only to the BC curriculum.*

Sequences and Series

1. Definitions
   a. Def'n: A sequence is a function whose domain is the set of positive integers. (The numbers in the range of the sequence are its elements or terms.)
   b. Def'n: A sequence $\{a_n\}$ has the limit L, written $\lim_{n\to\infty} a_n = L$, if to each number $\varepsilon > 0$, there corresponds an integer $N > 0$ such that $|a_n - L| < \varepsilon$ whenever $n \geq N$ (i.e. for any $\varepsilon > 0$ there exists an integer $N > 0$ such that all terms beyond the Nth in the sequence differ from L by less than ε).
   c. Def'n: A sequence that has a limit is said to be convergent, and to converge to that limit. A sequence that fails to have a limit is divergent.
2. Basic Theorems
   a. Thm: If $\lim_{x\to\infty} f(x) = L$ and f is defined for all positive integers, then also $\lim_{n\to\infty} f(n) = L$, where n is a positive integer.
   b. Thm: The limit of a convergent sequence is unique.
   c. Thm: If $\{a_n\}$and $\{b_n\}$ are convergent sequences and C is a constant, then
      i. $\lim_{n\to\infty} ca_n = c \lim_{n\to\infty} a_n$.
      ii. $\lim_{n\to\infty} (a_n \pm b_n) = \lim_{n\to\infty} a_n \pm \lim_{n\to\infty} b_n$.
      iii. $\lim_{n\to\infty} a_n b_n = \lim_{n\to\infty} a_n \lim_{n\to\infty} b_n$.
      iv. $\lim_{n\to\infty} (a_n/b_n) = \lim_{n\to\infty} a_n / \lim_{n\to\infty} b_n$ (provided $\lim_{n\to\infty} b_n \neq 0$).
3. Monotonic Sequences
   a. Def'n: A sequence is said to be monotonic and
      i. Increasing (or non-decreasing) if $a_n \leq a_{n+1}$ (i.e. $a_1 \leq a_2 \leq a_3 \leq \ldots$).
      ii. Strictly increasing if $a_n < a_{n+1}$ (i.e. $a_1 < a_2 < a_3 < \ldots$).
      iii. Decreasing (or non-increasing) if $a_n \geq a_{n+1}$ (i.e. $a_1 \geq a_2 \geq a_3 \geq \ldots$).
      iv. Strictly decreasing if $a_n > a_{n+1}$ (i.e. $a_1 > a_2 > a_3 > \ldots$).
4. Bounds
   a. Definitions
      i. "A" is called an upper bound of the sequence $\{a_n\}$ if $a_n \leq A$ for all positive integers n.
      ii. "B" is called a lower bound of the sequence $\{a_n\}$ if $a_n \geq B$ for all positive integers n.
      iii. A sequence $\{a_n\}$ is said to be bounded if there is a number "C" such that $|a_n| \leq C$ for all postive integers n. (Equivalently, $\{a_n\}$ is bounded if and only if it has an upper bound and a lower bound.)
   b. Axiom of completeness: Every set of real numbers which has a lower bound has a greatest lower bound (*glb*), and every set of real numbers which has an upper bound has a least upper bound (*lub*).
   c. Thm: A monotonic sequence is convergent if and only if it is bounded. (Note: For all sequences $\{a_n\}$ convergence implies boundedness, but not conversely.)
   d. More Theorems
      i. If $\{a_n\}$ is a monotonically increasing sequence which has an upper bound, then the sequence is convergent, its limit being the *lub* of the numbers $\{a_n\}$.

ii. If $\{a_n\}$ is a monotonically decreasing sequence which has a lower bound then the sequence is convergent, its limit being the *glb* of the numbers $\{a_n\}$.

Infinite Series

1. Definitions and Basic Theorems

a. Def'n: If $\{a_n\}$ is a sequence, then an expression of the form $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \ldots + a_m + \ldots$ is called an infinite series.

b. Def'n: Let $\sum_{n=1}^{\infty} a_n$ be an infinite series, and let $\{S_n\}$ where $S_n = a_1 + a_2 + a_3 + \ldots + a_n$ be the sequence of the partial sums of the series. If $\lim_{n\to\infty} S_n$ exists and is equal to S, the series is said to be convergent and S is called the sum of the infinite series $\sum_{n=1}^{\infty} a_n$. If $\lim_{n\to\infty} S_n$ fails to exist, the series is divergent and has no sum (If series $\sum_{n=1}^{\infty} a_n$ has a sum, we write $S = \sum_{n=1}^{\infty} a_n$).

c. Thm: (necessary condition for convergence) If the infinite series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n\to\infty} a_n = 0$. (Note: converse is not true.)

d. Corollary: If the nth term of an infinite series $\sum_{n=1}^{\infty} a_n$ does not approach 0 as $n \to \infty$, the series is divergent.

e. Thm: The two infinite series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=k}^{\infty} a_n$, for any positive integral k, both either converge or diverge.

f. Thm: If "c" is any non-zero constant and $\Sigma\, a_n$ converges, then so does $\Sigma\, ca_n$, and its limit is $c\Sigma\, a_n$. If $\Sigma\, a_n$ diverges, then so does $\Sigma\, ca_n$.

g. Thm: If $\Sigma\, a_n$ and $\Sigma\, b_n$ are convergent series, then $\Sigma\,(a_n + b_n)$ and $\Sigma\,(a_n - b_n)$ are also convergent, and $\Sigma\,(a_n \pm b_n) = \Sigma\, a_n \pm \Sigma\, b_n$.

h. Thm: The terms of a convergent series may be grouped in any way and the new series will converge to the same limit as the original series.

i. Thm: let $\{S_n\}$ be the sequence of partial sums of $\Sigma\, a_n$, then for every $\varepsilon > 0$ there exists a number N such that $|S_n - S_m| < \varepsilon$ whenever $n, m > N$.

2. Some Important Series

a. The Geometric Series: Σr^n is convergent for $|r| < 1$ with sum $= a/(1 - r)$ and divergent for all other values of r.

b. The Harmonic Series: $\Sigma\, 1/n$ is divergent.

c. The Hyperharmonic series $\Sigma 1/(n^k)$ is convergent for $k > 1$, and divergent for $k \le 1$.
d. The Telescopic Series $\Sigma\ 1/(n(n + 1))$ is convergent.
e. The series $\Sigma\ 1/n!$ is convergent.
f. Polynomial Test: If $P(n)$ and $Q(n)$ are polynomials of pth and qth degree respectively, then $\Sigma\ P(n)/Q(n)$ converges if $q > p + 1$ and diverges if $q \le p + 1$.

3. Tests For Convergence of Series of Positive Term
   a. Def'n: Series of positive terms only will be called positive series.
   b. Thm: A necessary and sufficient condition for an infinite positive series to be convergent is that its sequence of partial sums has an upper bound.
   c. Thm: (Comparison Test) Let Σa_n, Σb_n, Σc_n be positive series.
      i. If Σb_n is convergent and $a_n \le b_n$ (for all n), then Σa_n is also convergent.
      ii. If Σc_n is divergent and $a_n \ge c_n$ (for all n), then Σa_n is also divergent.
   d. Thm: (Integral Test) If f is positive, continuous, and decreasing for $x \ge 1$ and $a_n = f(n)$, then $\sum_{n=1}^{\infty} a_n$ and $\int_1^{\infty} f(x)\, dx$ either both converge or both diverge. If the series coverges to S, then the remainder $R_N = S - S_N$ is bounded by $0 \le R_N \le \int_N^{\infty} f(x)\, dx$.
   e. Thm: (Ratio Test) let Σa_n be a positive series, and let $\lim_{n\to\infty} (a_n + 1)/a_n = L$.
      i. If $L < 1$, the series converges.
      ii. If $L > 1$, the series diverges.
      iii. If $L = 1$, the test fails.

4. Alternating Series and Absolute Convergence
   a. Def'n: An infinite series of the form $\sum_{k=1}^{\infty} (-1)^{k+1} (a_k) = a_1 - a_2 + a_3 - \ldots$ where $a_k > 0$ (for all k), is called an alternating series.
   b. Thm: The alternating series $a_1 - a_2 + a_3 - \ldots$ converges if $0 < a_{n+1} < a_n$ (for all n) and $\lim_{n\to\infty} a_n = 0$.
   c. Thm: If $\sum_{k=1}^{\infty} (-1)^{k+1} (a_k) = a_1 - a_2 + a_3 - \ldots$ is an alternating series such that $0 < a_{k+1} < a_k$ and $\lim_{k\to\infty} a_k = 0$, the error made by using the sum of the first k terms as an approximation for the sum of the series is less than a_{k+1}. The absolute value of the first neglected term, i.e.

$$\left|\sum_{n=1}^{\infty}(-1)^{n+1}a_n - \sum_{n=k}^{\infty}(-1)^{n+1}a_n\right| < a_{k+1}$$

   d. Def'n:
      1. The infinite series Σa_n converges absolutely if $\Sigma |a_n|$ converges.
      2. If a series is convergent but not absolutely convergent, it is said to be conditionally convergent.
   e. Thm: If a series converges absolutely, then it also converges.
   f. Thm: For any series of nonzero terms Σa_n, let $\lim_{n\to\infty} |(a_{n+1})/a_n| = L$.

1. If $L < 1$, series converges absolutely.
2. If $L > 1$, series diverges.

g. Thm: If a series is absolutely convergent, its terms may be rearranged without affecting the absolute convergence of the series or its sum.

h. Thm: If $\Sigma\, a_n$ and $\Sigma\, b_n$ are absolutely convergent series with sums A and B, respectively, then the series $a_1b_1 + (a_1b_2 + a_2b_1) + (a_1b_3 + a_2b_2 + a_3b_1) + \ldots$ converges absolutely and its sum is AB.

Power Series

1. Definitions and Basic Theorems

a. Def'n: If $\{a_n\}$ is a sequence of constants, the expression $\sum_{n=0}^{\infty} a_nx^n = a_0 + a_1x + a_2x^2 + a_3x^3 \ldots$ is called a power series in x.

b. If the power series $\Sigma\, a_nx^n$ converges for a number $x_1 \neq 0$ then it converges absolutely for all number x such that $|x| < |x_1|$.

c. If the power series diverges for a number x_2, then it diverges for all numbers x such that $|x| > |x_2|$.

d. Let $\Sigma\, a_nx^n$ be a given power series. Then exactly one of the following conditions holds:
   i. The series converges only when $x = 0$.
   ii. The series is absolutely convergent for all values of x.
   iii. There exist a number $r > 0$ (called the radius of convergence) such that the series is absolutely convergent for all values of x for which $|x| < r$ and is divergent for all values of x for which $|x| > r$ (converges possibly at one or both endpoints r, $-r$).

e. Def'n: The set of all values for which a power series is convergent is called the interval of convergence of the power series.

f. Similarly, if a_n is a sequence we define the expression $\sum_{n=0}^{\infty} a_n(x - b)^n = a_0 + a_1(x - b) + a_2(x - b)^2 + a_3(x - b)^3 + \ldots$ to be a power series in $(x - b)$.

g. The conditions in the above theorem now become:
   i. The series converges only when $x = b$.
   ii. The series is absolutely convergent for all values of x.
   iii. There exists a number $r > 0$ such that the series is absolutely convergent for all values of x for which $|x - b| < r$ and is divergent for all values of x for which $|x - b| > r$ (converges possibly at one or both endpoints $b - r$, $b + r$).

2. More on Power Series

a. Def'n: Since the power series $\Sigma\, a_n(x - b)^n$ has a unique sum at each point in its interval of convergence, a function f is defined by the power series, i.e. $f(x) = \Sigma\, a_n(x - b)^n$. The domain of f is the interval of convergence of the power series, and the values of f at any point x in this domain is the sum of the power series at this point.

b. Def'n: The derived series of the power series $\Sigma\, a_n(x - b)^n$ is $\Sigma\, na_n(x - b)^{n-1} = a_1 + 2a_2(x - b) + 3a_3(x - b)^2 + \ldots + na_n(x - b)^{n-1} + \ldots$, i.e., the series formed by taking the derivates of the terms of the original series.

c. Functions defined by power series behave very much like polynomials as is indicated by the following theorem. Thm: Let $\Sigma\, a_n(x - b)^n$ be a power series in $(x - b)$ and let f be defined by $f(x) = \Sigma\, a_n(x - b)^n$, where x is any number in the interval of convergence of the series.
   i. f is continuous at every point in the interval of convergence.

ii. The derived series of the power series is equal to the derivative of the function defined by the power series at every point in the inteval of convergence, except possibly at the endpoints, i.e. $f'(x) = \sum D_x [a_n(x - b)^n] = \sum na_n(x - b)^{n-1}$.

iii. The series obtained by integrating the terms of the given series converge to $\int_b^x f(t)\,dt$ for each x within the interval of convergence of the original power series, except possibly at the endpoints, i.e. $\int_b^x f(t)\,dt = \sum_{n=0}^{\infty}\int_b^x a_n(t-b)^n\,dt = \sum_{n=0}^{\infty} a_n \frac{(x-b)^{n+1}}{n+1} = a_0(x-b) + a_1(x-b)^2 + \cdots$

(Note: It follows from ii above that a function that can be represented by a power series is infinitely differentiable and all derived series have the same radius of convergence (not necessarily same interval).)

3. Taylor Series and Taylor's Theorem
   a. If a function f is infinitely differentiable, the series $\sum_{n=0}^{\infty}\frac{f^{(n)}(b)(x-b)^n}{n!} = f(b) + f'(b)(x-b) + \frac{f''(b)(x-b)^2}{2!} + \frac{f'''(b)(x-b)^3}{3!} + \cdots$ is called the Taylor series in $(x - b)$ of the function.
   b. Consider any power series in $(x - b)$, $a_0 + a_1(x - b) + a_2(x - b)^2 + \ldots$ and denote its interval of convergence by $(b - r, b + r)$, where $0 \leq r < \infty$. Denote by the function defined by $f(x) = a_0 + a_1(x - b) + a_2(x - b)^2 + \ldots + a_n(x - b)^n + \ldots$ where x is any number in $(b - r, b + r)$. Then the power series is the Taylor's series in $(x - b)$ of the function f, namely $f(x) = f(b) + f'(b)(x - b) + f''(b)(x - b)^2 / 2! + \ldots + f^{(n)}(b)(x - b)^n / n! + \ldots$ where $b - r < x < b + r$ (i.e. $a_n = f^{(n)}(b) / n!$).
   c. The Taylor series in x is called a Maclaurin series, $f(0) + f'(0)x + f''(0)x^2 / 2! + f'''(0)x^3 / 3! + \ldots + f^{(n)}(0)x^n / n! + \ldots$ A Taylor series in $(x - b)$ of a function f is sometimes called the Taylor expansion of f about the point b.
   d. Taylor's Theorem: Let f be a function which is defined in some open interval $(b - r, b + r)$, $0 \leq r < \infty$, and whose first $n + 1$ derivatives exist and are continuous throughout the interval. Then for all numbers x in $(b - r, b + r)$, $f(x) = f(b) + f'(b)(x - b) + f''(b)(x - b)^2 / 2! + \ldots + f^{(n)}(b)(x - b)^n / n! + R_n(x)$ where $R_n(x) = f^{(n+1)}(\xi)(x - b)^{n+1} / (n + 1)!$ (Lagrange form of the remainder) where $b < \xi < x$.
   e. Other forms of the remainder are:

      i. Integral form: $R_n(x) = 1/n! \int_b^x (x - t)^n f^{(n+1)}(t)\,dt$.

      ii. Cauchy's form: $R_n(x) = f^{(n+1)}(\xi)(x - \xi)^n(x - b) / n!$, where ξ is some number between b and x.

   f. Thm: Since $f(x) - \sum_{k=0}^{n} f^{(k)}(b)(x - b)^k / k! = R_n(x)$, it follows that if f is a function having derivatives of all orders in some interval $(b - r, b + r)$. A necessary and sufficient condition that the Taylor's series $\sum_{n=0}^{\infty} f^{(n)}(b)(x - b)^n / n!$ represents the function f at all points in the interval is that, whenever $|x - b| < r$, $\lim_{n\to\infty} R_n(x) = 0$, where $R_n(x)$ is the remainder after $n + 1$ terms in Taylor's formula.

Important Series Expansions

1. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + \frac{(-1)^{n-1}x^{2n-1}}{(2n-1)!} + \cdots -\infty < x < \infty.$

2. $\sin x = \sin a + (x-a)\cos a - \frac{(x-a)^2 \cos a}{2!} - \frac{(x-a)^3 \cos a}{3!} + \cdots -\infty < x < \infty.$

3. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + \frac{(-1)^{n-1}x^{2n-2}}{(2n-2)!} + \cdots -\infty < x < \infty.$

4. $\cos x = \cos a - (x-a)\sin a - \frac{(x-a)^2 \cos a}{2!} + \frac{(x-a)^3 \sin a}{3!} + \cdots -\infty < x < \infty.$

5. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots -\infty < x < \infty.$

6. $\frac{1}{1-x} = 1 - x + x^2 - x^3 + \cdots -1 < x < 1.$

7. $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots -1 \le x < 1.$

8. $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots -1 < x < 1.$

9. $\ln(1-x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \cdots -1 < x < 1.$

10. $\ln(x) = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \cdots 0 < x \le 2.$

11. $\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \cdots -1 < x < 1.$

12. $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots -1 \le x < 1.$

13. The General Binomial Theorem: (for all real m, $-1 < x < 1$)

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \cdots + \frac{m(m-1)\cdots(m-n+1)}{n!}x^n + \cdots$$

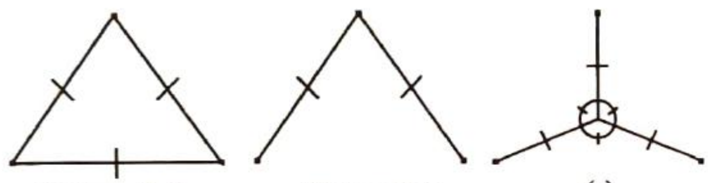
Solving Steiner's Problem

BY GARY LI

VIRTUALLY EVERYONE IS familiar with Euclid's first postulate which states that the shortest distance between any two points is a straight line, but what is the minimum distance - that which requires the least energy to travel - between any number of points? This is the Steiner Problem which is named after J. Steiner who first investigated it in the early 19th century.

To begin, let us consider three points arranged as the vertices of an equilateral triangle of perimeter 3 (See Figure 1a). One may first connect all the points giving a length of 3 but this can by shortened by eliminating one side (See Fig. 1b) because they will still be connected. Now we have a length of 2 units but is this the shortest path? It would take too long to find out by simply doodling so we shall first employ an analog (or physical) solution.

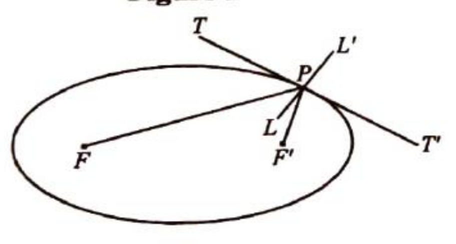
Figure 1



(a) Length 3 (b) Length 2 (c)

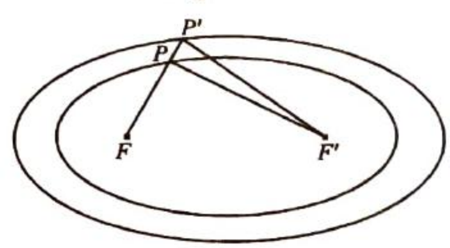
The analog solution merely involves constructing two parallel transparent panes with a number of pins equal to the number of relevant points and in this case, three, pins holding them together such that they form the vertices of an equilateral triangle and then dipping this into a container of a soap solution. Given that soap films have a tendency to minimize their surface area (and thus looking at it from above length as well) the film of soap formed as a result reveals the minimum path linking the three points (See Fig. 1c) and in general the length of the minimum path is equal to $s\sqrt{3}$ provided all three points are equidistant and s is the distance between any two points. Therefore in this case, since s is equal to one, the path has a length of $\sqrt{3}$ which is certainly the shortest.

Figure 2



An ellipse with foci F and F'

Figure 3



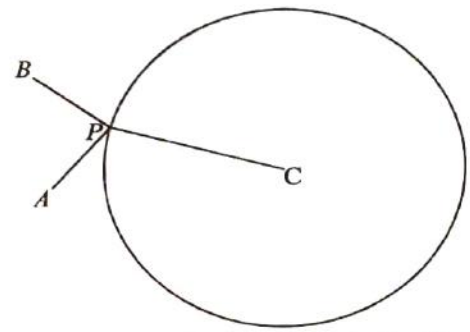
Confocal Ellipses

One can prove this with some analysis. An ellipse is such that given foci F and F' (See Figure 2) and any point P on the ellipse, $FP + F'P = C$ is a constant. FP and $F'P$ also form equal angles with the line tangent to the ellipse at P so $\angle TPF = \angle TPF'$, $FPL = F'PL$, and $FLL' = F'LL'$. Given confocal ellipses (See Figure 3) with foci F and F' triangle PP'F' the sum of two sides must be greater than the remaining one therefore $PP' + P'F' > PF'$ so by adding FP to both sides, $FP + PP' + P'F' > FP + PF'$ and since $FP + PP' = FP'$, $FP' + P'F' > FP + PF'$. In considering the minimum path between three points let P be the point of intersection between the straight lines of minimum path connecting three points, A, B, and C (one can eliminate any curves since they can be replaced by straight lines). Let us first assume P does not coincide with one of the points (thus A and B would not inside circle C).

Now in a circle with center C and radius CP with Z any point and assuming P is such that $AQ + BQ + CQ$ is the smallest possible value. Therefore any point, P', on the circle must be such that $AP' + BP' + CP' \geq AP + BP + CP$ since CP' and CP are both radii $AP' + BP' \geq AP + BP$ thus $AP' + BP'$ must

foundations of knowledge in mathematics. — Roger Bacon

Figure 4

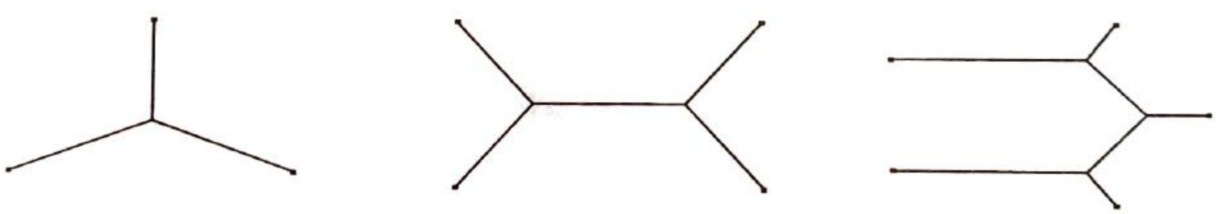


Three Points A, B, C, with a circle of radius CP

be minimal when the path is minimal. If one adds a set of confocal ellipses with foci A and B the ellipse that intersects the circle will have a minimal $AP'+BP'$. Intersection in two different points would result in a greater value of $AP' + BP'$ therefore P must be the point of intersection of the ellipse with the circle therefore $\angle APC = \angle BPC$ since PC is normal or perpendicular to the tangent at P. Applying this to the circle of center A and radius AP, $\angle APB = \angle APC$ therefore angle $\angle APC = \angle APB = \angle BPC$. Since the sum of the angles is equal to 360° each of the three must be equal to 120° thus P is the point that subtends a 120° angle with any pair of lines from A,B,and C provided triangle ABC does not have angles greater than 120° at which point the minimal path would be the two sides of the triangle that are adjacent to the obtuse angle.

Let us now consider what would happen if either A or B was inside circle C but with P not coinciding with a vertex. If one of the vertices, perhaps A, is inside the circle therefore $AP+BP \geq AB$ (since P is the point of intersection between lines from the points constituting the minimum path) and since the sum of two sides must exceed or equal the third thus $CP \geq AC$ (since AC has a length at least equal to radius CP) therefore $AP + BP + CP \geq AB + AB$. This would lead one to conclude that the shortest path is when P coincides with A but this cannot be because we have said P cannot coincide with a vertex. Therefore A and B must be outside the circle.

Minimum lengths for:



(a) Three Points (b) Four Points (c) Five Points

Figure 5

In sum the minimum path between three points A, B, and C consists of line drawn from each point to a point, P, such that the angle of intersection between them is 120°. If one then considers a larger number of points (See Figure 5) three generalization for the solutions of Steiner's Problem becomes evident. One that the angle of intersection between surfaces is either 109° or 120°. Two, that the number of intersection cannot exceed (n-2) where n is the number of points being considered. Third it seems that as the number of points treated are increased the path grows closer in similarity to a hexagon (See Figure 6) tessellation (covering without overlap or gaps) of a plane though many lines are missing. From this one can find that the number of points that constitute the minimum path between any number of points on a two dimensional plane, provided that every single point is set at equal distance to each of its surrounding neighbors, n, is equal to $6h - ei + e$ (empirically derived and unproved) where n is the number of hexagons (which can be found by adding as many lines between any two points that complete a hexagon as is possible), i is the number of common sides between any two hexagons, and e is the number of points not part of any hexagon. Thus the minimum length is equal to $6l\left[(n+2i-e)/6\right]$ and the sum of the lengths of all the extra points to their closest neighbors and minus hl where l is the common distance between points. This can also be applied to a non-conformist set of points provided they still form the same number of hexagons as a conformist set with same number of points and if they do not then the by connecting each point to its closest neighbor and then adding up all the lengths one can find the minimum path. In drawing a solution to a Steiner's Problem one need only connect all the points to form as many hexagons as possible then eliminate all unneeded lines.

One final note there is another method other than physically solving a Steiner's Problem in which one can use the Euler-Lagrange equation. But we must first consider the process involved for determining the shortest distance between two points (Skip to the conclusion if calculus bores you). Let two points (see Figure 7). have an element of distance ds, along a varied path which is

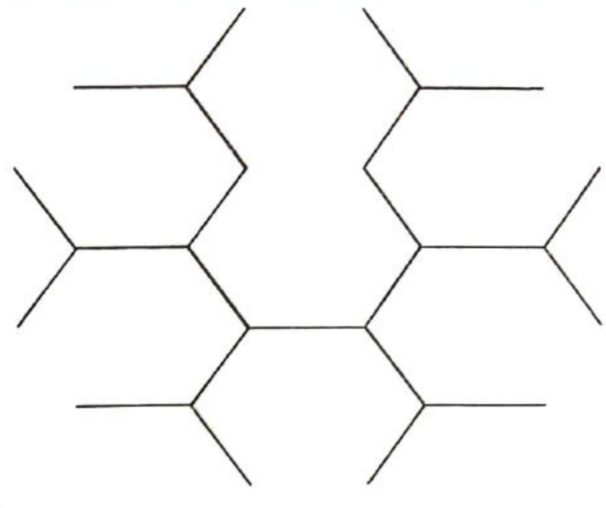


Figure 6

equal to $[(dx)^2+(dy)^2]^{1/2}$. The total length of the varied path s can be given by $s=\int_{x_1}^{x_2}[1+(dy/dx)^2]^{1/2}dx$

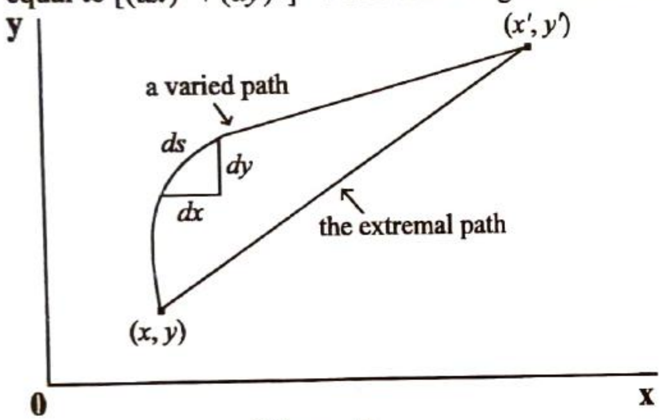


Figure 7

and knowing the Euler-Lagrange equation $\frac{d}{dx}\left(\frac{\partial f}{\partial y_x}\right)=0$ thus $\frac{\partial f}{\partial y_x}=K$ where K is a constant.

By substituting $\frac{\partial f}{\partial y_x}\left(1+y_x{}^2\right)^{1/2}=K$ thus $y_x\left(1+y_x^2\right)^{-1/2}=K$. The left side is a function of y_x therefore y_x must be a constant so let $y_x=m$ so that $m\left(1+m^2\right)^{-1/2}=K$ thus $\frac{dy}{dx}=m$ therefore $y=mx+c$ which is a line thus the shortest distance between two points is a straight line. If such an effort is required for merely proving that the shortest distance between two points is a straight line the effort for a more significant number of points would be much harder, though challenging, thereby rendering its impractical for a solution to a Steiner's Problem.

To conclude there is presently no mathematically general solution to Steiner's Problem but by the existence of a minimizing tendency in soap films does enable us to acquire an analog solution and could prove to play an important role in the search for a general analytic solution, which will probably involve a mathematical model.

References

Almgren, F.J.Jr. " The Geometry of Soap Films and Soap Bubbles," *Scientific Amerian*, July 1976.
Boys, C. V. *Soap Bubbles, their colours and the forces which mold them*, New York: Dover , 1959.
Isenberg, Cyril, *The Science of Soap Film and Soap Bubbles*, New York: Dover, 1992.
Newman, J.R. *The World of Mathematics Vol.II*, Simon and Schuster, 1956.
Rouleau, Raynald "Are You Smarter than a Soap Bubble?" *21st Century*, Spring 1993.

I have hardly ever known a mathematician who was capable of reasoning. — Plato

Eratosthenes

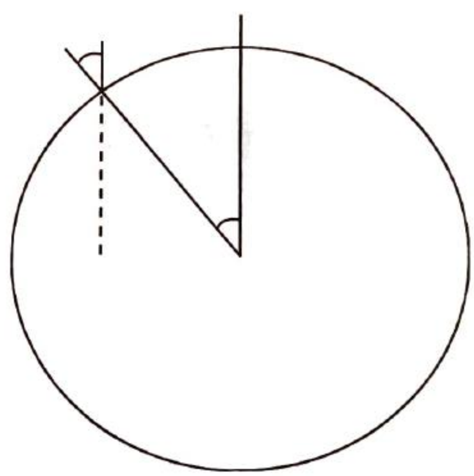
BY TOMMY WU

THE SIEVE OF Eratosthenes allows us to find prime number from a set of natural numbers. Prime numbers are numbers that have only two factors-one and itself. On the other hand, composite numbers have more than two factors. (The number one is neither prime nor composite. It has only one factor.)

Eratosthenes reasoned that if a number, x, was a multiple of another number, y, then x could not be prime number because y would be a factor of x. He used this to work out the Sieve.

To create the Sieve , one must first list the natural numbers. Cross out the number one, because it is neither prime nor composite. Then cross out all multiples of 2, except for 2. Cross out all multiples of 3, except for 3. Cross out all multiples of 5, except for 5. Continue to cross out all multiples of n except for n, and only prime numbers will be left. The non-prime numbers "flow" out and the prime numbers are left behind on a "sieve."

Eratosthenes was perhaps more famous for discovering a method to measure the circumference of the earth. Eratosthenes discovered that at the summer solstice, the sun shone at a ninety-degree angle in Syene (now Aswan) at noon. He knew this because the reflection of the sun could be seen in a deep well. At the same time, he found that in Alexandria, the sun shone at an altitude of seven degrees off the perpendicular. Since the light rays from the sun were assumed to be parallel, the difference must have been caused by the earth's roundness. Looking at the diagram, you will see that this angle difference is equal to the arc of the earth. (If two parallel lines are cut by a transversal, the corresponding angles are equal in measure.)



This arc measures seven degrees, which is about 1/50 of the earth's circumference of 360 degrees. Eratosthenes then reasoned that the distance between Syene and Alexandria must be 1/50 of the length of the circumference of the earth. Once he obtained the distance between the cities he was able to calculate the earth's circumference.

Eratosthenes used a unit called the stade, and mathematicians are unsure of what the unit means. However, they know that his method works because the earth's circumference is approximately fifty times the distance between Syene (Aswan) and Alexandria.

To help understand Eratosthenes' method better, it can be thought as this proportion

$$\frac{\text{Difference in sun's altitude}}{360^\circ} = \frac{\text{Distance between cities}}{\text{Earth's circumference}}$$

It may be true that people who are merely mathematicians have certain specific shortcomings; however, that is not

Using Eratosthenes' measurements :

$$\frac{7^\circ}{360^\circ} = \frac{5{,}000 \text{ stades}}{250{,}000 \text{ stades}}$$

Using our own measurements, let n equal the earth's circumference :

$$\frac{7^\circ}{360^\circ} = \frac{840 \text{ kilometers}}{n}$$

$$7n = 302{,}400$$

$$\frac{7n}{7} = \frac{302{,}400}{7}$$

$$n = 43{,}200.$$

Or simply multiply the distance between the cities by 50, as Eratosthenes did.

$$840 \times 50 = 42{,}000.$$

In any case, the final results are close to the actual circumference, which is 40,008 kilometers.

References

Daley, Robert B., Higham, W. John, and Matthias, George F., *Earth Science : A Study of a Changing Planet.* New Jersey: 1978.

the fault of mathematics, but is true of every exclusive occupation. — Carl Friedrich Gauss

The Mathematics of Music

BY HELEN HONG

THROUGHOUT MATHEMATICAL HISTORY, it has been proven and accepted that there exists a clear relationship between the seemingly analytical mathematics and creative music. Upon closer consideration one can observe that working with mathematics requires high levels of creativity while music can be easily analyzed by its purely technical properties. Precipitating from the study of a correlation between these two subjects is a surprising discovery made by Richard F. Voss, a physicist from Minnesota who works at the Thomas J. Watson Research Center of the International Business Machines Corporation. Voss encountered fractal music, which may be studied and generated solely by the use of mathematics.

When one tries playing a recording of typical sounds, such as music or voice, at different speeds, one may expect the character of the sound to change accordingly. Scaling noises , however, behave in an unlikely manner. If you play a recording of such a sound at a different speed, you only have to adjust its volume in order to make it sound exactly as it did before. The simplest example of a scaling noise is what in electronics and information theory is called white noise (or "Johnson noise"). White noise is a colorless hiss that is just as dull at any speed that it is played at. The most commonly encountered white noise is the thermal noise made by the random motions of electrons through an electrical resistance. Thermal noise causes most of the static in a radio or amplifier and the "snow" on radar and television screens when there is no input.

One can compose a random "white tune", one with no correlation between any two notes, by using randomizers such as dice or spinners. Taking a scale of one octave of seven white keys on a piano (do, re, mi, fa, so, la, and ti), construct a spinner similar to the one shown in the figure below. The arc lengths assigned to the sectors may be completely arbitrary. To produce a "white melody" simply spin the spinner as often as you like, recording each chosen note. The result is a totally uncorelated sequence. It is also possible to divide the circle into more parts and let the spinner select notes that range over the entire piano keyboard, black keys as well as white.

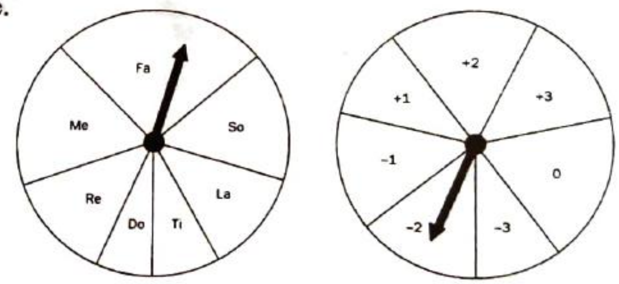


FIGURE 1 Spinners for white music (left) and brown music (right)

A more complicated kind of scaling noise is one that is called "Brownian noise" because it is characteristic of Brownian motion, the random motions of tiny particles suspended in a liquid and buffeted by the thermal agitation of molecules. Each particle executes a three-dimensional "random walk," the positions in which form a highly correlated sequence. A random walk can be described as a sequence of steps whose size and direction are determined by chance. In a simple, one-dimensional version of a random walk, a person flips a coin and takes one step forward if the result is heads and one step backward if the result is tails.

When tones fluctuate in this fashion, they are said to create Brownian music or brown music. We can produce it in almost the same way as we did before with the white music. The circle is divided into seven parts, of any size, and the regions are labeled to represent intervals between successive tones. On the spinner shown, plus means a step up the scale of one, two, or three notes and minus means a step down of the same intervals.

Begin the melody on the piano's middle C, then use the spinner to generate a linear random walk up and down the keyboard. The tune will wander here and there, and will eventually wander off the keyboard. The ends of the keyboards may represent "absorbing barriers" where the tune ends when encountered.

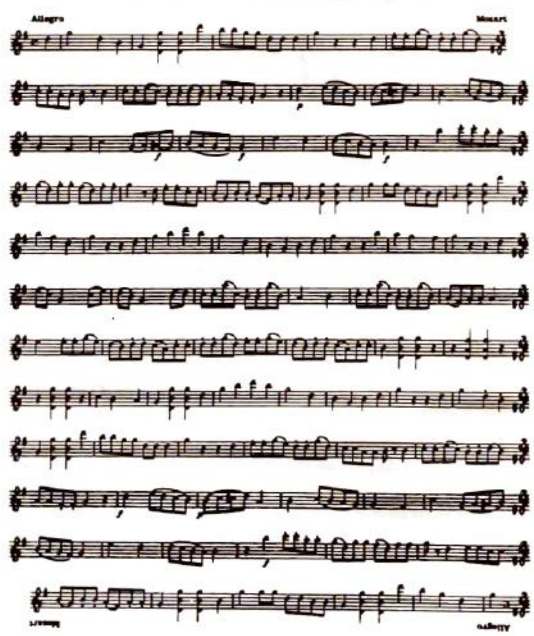
It is pathetic to watch the endless efforts—equipped with microscopy and chemistry, with mathematics and electronics—to reproduce a single violin of the kind the half-literate Stradivarius turned out as a matter of routine more than two hundred years ago. — Michael Polanyi

Voss's insight was to compromise between white and brown input by selecting a scaling noise precisely halfway between. In spectral terminology it is called $1/f$ noise. White noise has a spectral density of $1/f^0$, brownian noise a spectral density of $1/f^2$. In "one-over-f" noise the exponent of f is 1 or very close to 1. Tunes based on $1/f$ noise are moderately correlated throughout runs of any size. It turns out that almost every listener agrees that such music is much more pleasing to listen to than white or brown music.

In electronics $1/f$ noise is sometimes called flicker noise. Mandelbrot was the first to recognize how widespread $1/f$ noise is in nature, outside of physics, and how often one encounters other scaling fluctuations. For example, he discovered that the record of the annual flood levels of the Nile is a $1/f$ fluctuation. He also investigated how the curves that graph such fluctuations are related to "fractals," a term he invented. A scaling fractal can be defined roughly as any geometrical pattern (other than Euclidean lines, planes and surfaces) with the remarkable property that no matter how closely you inspect it, it always looks the same, just as a slowed or speeded scaling noise always sounds the same. Mande!brot coined the term fractal because he assigns to each of the curves a fractional dimensión greater than its topological dimensions. Thus, $1/f$ noise may also be termed as fractal music.

There is no doubt that music of almost every variety does exhibit $1/f$ fluctuations in its changes of pitch as well as in the changing loudness of its tones. Voss found this to be true of classical music, jazz and rock. He suspects it is true of all music.

One fascinating example of creativity and technicality intertwining math and music is a famous canon that was supposedly written by Mozart as a joke. In this instance the second melody is almost the same as the one you see taken backward and upside down. Thus, if the sheet of music is placed flat on a table, with two singers standing opposite each other, the singers can read from the same sheet as they harmonize.



Mozart's palindromic and invertible canon

Bibliography

Gardner, Martin. *Fractal Music, Hypercards and More...* New York: W.H.Freeman and Company, 1992.

Peterson, Ivars. *The Mathematical Tourist*. New York: W.H.Freeman and Company, 1988.

A Twisted Journey

BY BORIS KHENTOV

IT HAPPENED ON that remarkable day after school, when I had nothing to do. I was waiting for a friend in the library and I began playing with the graphic calculator that I had recently bought from the school.

"I wonder what this button does," I thought. As I pressed it, something strange began to happen. It felt like I was literally pulled into the calculator. In a split second, I found myself standing in the middle of nowhere. I looked up, and saw a line stretching up. I couldn't see it's end. There was one going down, left and right.

Slowly, it came to me. I was on the two-dimensional coordinate plane! I looked all around me. There was complete emptiness everywhere, except when I tried to look out of the plane. There, I could see the library ceiling. I took a few steps to the left, on which I assumed to be the x-axis. Then I walked back to the origin and sat down. I was confused and perplexed. Suddenly. my friend's face hovered over me. I waved frantically, but it looked like he couldn't see me. It seemed like he was punching buttons. He was, no doubt trying to graph something. I looked up into the second quadrant, just in time to see a rapidly approaching curve.

"Parabola coming through!" It hollered. "Out of the way kid!"

I had no time to move. The parabola's focus was of course the origin. In a split second, I was swept off my feet and tossed into the first quadrant. The parabola continued its journey upwards. Now, I was floating somewhere, unable to move myself. I could see the axes, but I couldn't get to them.

"How do I get out of here?" I screamed into the emptiness.

"The same way you got in." A voice answered.

"Who is there?" I inquired.

"I am the Graphmaster." The voice answered. "I guess you can call me the god of this plane."

"So how do I get back to my world?" I persisted.

"The same button that you pushed to get in, must be pressed again, and you must be on the origin when it is pressed."

"How do I get to the origin? I can't even move!"

"You'll have to use curved to help you. Oh, and your knowledge of coordinate geometry. But hurry....." His voice began to fade away.

"Wait! Don't go yet!" I screamed. "I still have to ask you...Oh well."

He was gone. I looked around. It seemed that I was about three units away from the origin. I looked towards the second quadrant and saw that a circle was forming. It got closer, and when it passed me, I grabbed on.

"Hello there!" said the circle.

"Quick!" I told him. "Get me to the origin?"

"Oh goodness, no! That's my center, and I can never go there. I have to stay exactly three units away from it. That's my equation, you know."

"Can't you make an exception just this time! Please?" I begged.

"Go against my formula?! Never!!!"

"Well then, do you know anyone who could get me to the origin?"

"You could try $y = x$. He is a nice guy, but so terribly straight! I'm fond of him though. He happens to be one of my diameters."

"How many do you have?" I asked.

"Oh, infinitely many. As long as a line is in the form of $y = mx$, it is my diameter."

"Good thing I didn't set this thing in the polar coordinate system." I thought. "I wouldn't have a clue about what this guy was talking about."

"So how do I find?" I asked out loud.

"Well..." he began, "NOOOOO!!!! My....My center! It's splitting ! I'm......I'm turning elliptic!!!!!! AAAARRRRGGG!!!!" Surely enough something weird was happening to Circle. He was bulging at the sides, and getting squished from the top.

"I know!" I exclaimed. "No longer are you $x^2+y^2=r^2$. Your formula is now $\frac{x^2}{a}+\frac{y^2}{b}=$!"

He didn't listen. It looked like he was devastated at no longer being a true circle anymore. I decided to leave him alone. I looked around to see if there were any other curves in sight. And there was a sine curve coming my way. As it came closer, I reached out and held on, leaving behind the distressed ex-circle.

"Where are you heading? And how did you get in this plane?" the sine curve asked.

"Long story." I replied. "I have to get to the origin."

"Sorry," he answered, "just passed that a few radians ago."

Suddenly I felt very sick. Up and down, and up and down and up and ....

"Can...can you let me ...off ..please?" I stammered.

"What's the matter?"he laughed. "Feeling a bit sine-sick? Ha, ha, ha! Don't mind my sine humor! Sure, I'll let you off at 10π." I slumped down on the x-axis as the sine curve passed me. It took me a while to come back to my senses. 10π? That's more than 30 units away from the origin! Stupid sine curve! Now I am even farther from the origin than I was before. Well, at least I am on the x-axis, so I can hike back. After some time, I heard a whistling sound.

"Yahooo! $x=25$ comin' through!" I heard someone yell.

In an instant, something hurled me up into the first quadrant again. I looked around, only to see a line rushing past me. I was stuck in mid-air again! Luckily I soon saw a line heading for me. I grabbed on as it passed me.

"Hi, who might you be?" I asked.

"I'm $y=x$. What do you want?"

"Can you get me to the origin?"

"I just passed it."

"Damn! Where are you going now?"

"Oh, I don't know. Towards infinity, I guess. Everything's fine by me, as long as y equals x."

"Infinity?! That could take forever!"

"If don't want to join me, you can take $y=2$ (to the xth power)."

He was right. Just above us was a curve that I could reach. I didn't bother to say "good-bye" to $y=x$, and jumped over to the curve.

As soon as I grabbed on, I felt that the curve was really slippery. I couldn't get a grip, and just slid down. To my horror, the curve was an enormous slide. On and on I slid, until I stumbled across the y-axis. The origin was just a unit away, but I missed it! I kept sliding into the second quadrant. After a while, the curve got so close to the x-axis, that I could get off. When I did, and looked towards the origin, I realized that the origin was at least 300 units away! In despair, I flopped down and fell asleep.

I was woken up with a jolt. When I opened my eyes, I saw that I was being carried by a line, and that the origin was getting closer.

"Who..who are you?" I asked, still startled.

"I am $y=0$. I will take you to the origin." He answered.

"Thank you...I think."I replied. Needless to say I was confused. How did he know where I was going?

"Here you go! Your stop!" The line said, and pushed me off.

I found myself right on the origin. "Finally!" I thought. Then I looked around cautiously. The last thing I needed now was a curve to throw me somewhere. I looked out of the plane. All I saw was the library ceiling. My friend was gone. Who was going to push that button now? I was stuck here forever!

"GRAPHMASTER!" I yelled at the top of my voice. "What do I do?!?!?!?!?!?!"

"You had the power to go home all along." replied the voice. "All you have to do is clap your heels together and say 'There's no place like home' three times."

"No!" I screamed."You're lying!!!! I can never go home! I can never go home! I am stuck in this stupid plane foreeeeeeeeeveeeeeeeeeerrrrr........................."

"Wake up!"

Someone was shaking me.

"Wake up!!!"

I looked up. I was sitting at the library table, with my calculator in front of me.

"I'm sorry I'm late. Looks like you fell asleep." My friend said. I ignored him. All I could do was stare at the screen. It was blank, except for one thing. $x = 0$.

The Strange and Mystical Pythagoreans

BY LINDA HONG

WHO IN THE world was Pythagoras? Most of us would picture him as an old, bent over, senile man no more than four feet tall— the traditional ancient philosopher. The fact is that anyone who knew him would say he was handsome and godlike, not to mention a great intellect. (Relax, girls. He's DEAD). He was born in Sidon, Phoenicia in 582 B.C. to Mnemarchus and Pythias. Some say he was a descendent of Zeus. Others say he was the son of Apollo, but that would imply adultery, so the former is more accepted.

Little Pythagoras was brought up in Samos and was educated in divine matters. Not surprisingly, he became very smart and was respected even by the eldest of the village, no matter how senile. When he was eighteen years old, he fled with his friend, Hermodamas from the growing tyranny of Polycrates. First he went to Pherecydes. Then to visit Anaxamander, and then finally stopped at Miletus to learn from Thales. (With all these long names, you'd think they would just call them Bob or Joe. Actually Pythagoras was nicknamed "the long haired Samian." He must have been an ape!)

Later he spent 22 years in Egypt to study astronomy, geometry, and all the mystic rites of the gods. Then he was taken to Babylon by Cambyses' soldiers. There he learned from the magi their solemn rites and perfect worship of the gods. He received the highest knowledge of the mathematical sciences. Pyth returned to Samos after another twelve years at the age of 56 and was considered even more handsome. (Talk about growing old gracefully!) There he acquired a pupil also named Pythagoras, but son of Eratocles. Every time the young boy learned something, Pyth would give him 3 obol as a wage. When the boy learned everything, he still wanted to learn more, even without pay! (If this method was used with every student, then we wouldn't have as many problems with attendance. We wish.)

Well this hairy Samian set up a school in Samos where many people came to see him. He also built another great school called *Magna Graecia* in Croton, Italy. He traveled far and wide to teach others. The people whom he taught were called the Pythagoreans.

They were taught not only math but morals, too. Pythagoras' moral teachings were much like the Ten Commandments of the Jews. (They even had a old hairy dude — Moses). Some Pythagorean principles were like those of Christianity. They included be friends to enemies, be kind to others, respect elders, don't be lazy, don't commit adultery, and don't misuse the gods' names in an oath. Pythagoras promoted patience, self control and sound-mindedness, sagacity and piety, and above all things, having mental culture was the best.

The Pythagoreans honored and revered Pythagoras greatly during his life and after his death. They secretly believed that there sere three kinds of rational living beings — the divine, the human, and Pythagoras. They believed in the "reminiscence of previous lives which souls lived through before entering the bodies in which they happened to be living" or, more simply put, reincarnation.

Strangely enough, these people worshipped numbers (not very different from Stuyvesant students, I believe). Actually, they only worshipped the numbers of the decad (1-10) because these numbers are both "pure and free from material change." They are also "the most perfect and important numbers" because they are the closest to 1, the source of numbers. Life and a telepathic form of consciousness were ascribed to these numbers. Each number had a symbolic meaning.

One is the supreme entity representing the mind. It is equivalent to Zeus or Apollo, which meant it is also the creator and the central fire or hearth of the universe. All the planets revolve around it, making music at the same time, not that it's the sun. The sun only reflects One's light. (Huh?) One was considered male (naturally by men, of course) and it produced primal motion or the dyad, which is Two. Two, the female, and One then produced Three, the first number and symbol of the cosmos. One and Two are not numbers at all; they are the creators of numbers.

Geometrically One represents a point; the Dyad, a line; Three, a plane; and, obviously Four represents solid bodies. Four was also the key-bearer of nature. Five is the change of quality, light and justice, and the

Mathematics is the only science where one never knows what one is talking about nor whether what is said is true.
— Bertrand Russell

smallest extremity of vitality. (Boy, these people must have been really twisted.) Six, since it is the first perfect number, is considered harmony, the perfection of parts, and Venus herself. The Pythagoreans believed it was the only number adapted to the soul, too. (See, I told you they were twisted.) Seven is Fortune, opportunity, and voice. The gods that Seven represent are Minerva and Tritogenia. Eight is love and friendship. Nine (were almost finished!) is the horizon and freedom from strife and similitude and it also represents a whole bunch of gods.

The decad itself was called heaven, fate, eternity, strength faith, necessity, and the world. Infinity was thought to be evil because it is so far from One. We can also conclude that negation was considered evil because Pythagoras never dealt with it. (Talk about paranoia!) As for Zero, well, it's a complete zero.

Bibliography

Gorman, Peter. *Pythagoras, a Life*. Boston:Routledge & K.Paul, 1979.
Iamblidius. *On the Pythagorean Life*. Atlanta, Ga.:Scolars Press,1991.
Taylor, Thomas. *The Theoretic Arithmetic of the Pythagoreans*. New York: Samuel Weiser,1975.

Measure two quarts for each lady.

HONEST JOHN says: "What I don't know about milk is scarcely worth mentioning," but he was flabbergasted one day when each of two ladies asked him for two quarts of milk. One lady had a five-quart pail and the other had a four-quart pail. John had only two ten-gallon cans, each full of milk. How did he measure out exactly two quarts of milk for each lady?

It is a juggling trick pure and simple, devoid of trick or device, but it calls for much cleverness to get two quarts of milk into those two pails without making use of any receptacles other than the two pails and the two full cans.

Answer

Call one ten-gallon milk can A and the other B, then proceed as follows:
Fill 5 pail from can A.
Fill 4 pail from 5 pail, leaving 1 quart in 5 pail.
Empty 4 pail into can A.
Pour the quart from 5 pail into 4 pail.
Fill 5 pail from can A.
Fill 4 pail from 5 pail, leaving 2 quarts in 5 pail.
Empty 4 pail into can A.
Fill 4 pail from can B.

Psychology cannot experiment with men, and there is no apparatus for this purpose. So much the more carefully must

Simplifying Complex Math for Children

BY LINDA LAU

THROUGH A SERIES of simple games, Mike Fellows, a theoretical computer scientist at the University of Victoria in Canada, and various mathematical researchers are introducing kids to some concepts of computer science and mathematics. He spray painted the grass at the elementary school to prepare for a math lesson on the workings of computer algorithms. It showed how a machine can take a set of numbers, which are not in order, and order them after a number of comparisons between pairs of numbers. Each child would carry a card with a number on it. They enter as " input " at one end, and move along the lines on the grass until they reach a " comparator node." At this point, they would compare cards, and whoever has a smaller number goes to the right, while the child with a larger value goes to the left. Eventually, they would be ordered. If this network is drawn properly, the kids would exit the networks in order, no matter how they entered. He is also working with Neal Koblitz, from the University of Washington in Seattle, to introduce cryptographic ideas to kids. But their system, Kid Krypto, has a few problems.

Paul Sally, a mathematician at the University of Chicago, has a few problems which interest people at all levels of mathematics. He is acknowledging elementary school teachers about some of them. One of his favorites is a number game on squares. Start with a counting number at each corner of a square. Write the difference of the two consecutive corners at the midpoints of the four sides. These midpoints are now new corners next to each other at their midpoints again. Draw a new square with the midpoints as corners again. Keep on doing this and the midpoints of the final square will be zero.

These mathematicians are all interested in teaching math to small children. Other mathematicians are holding summer work shops for high school math teachers. The National Council of Teachers of Mathematics developed a new set of standards which stresses problem solving and exploration of math patterns and relationships. The mathematical researchers stepped in to provide young children with just that.

MATH TEAM 1993-1994

BY LEO VOLFTSUN AND ARIF-US SAMAD

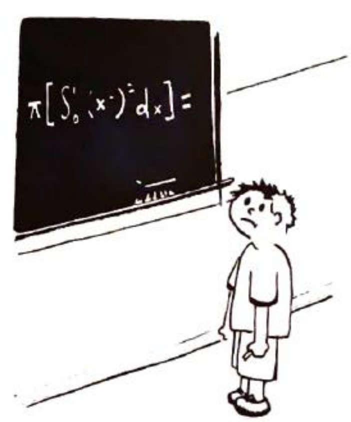
THIS YEAR AGAIN, the Math Team was the same old thing. Every morning, approximately 200 hardy souls with the idea " Math is #1" arrived and took their place in one of the Math Team rooms commanded by Leo Volftsun, Felix Liu, Il-Seung Cho, Helen Hong and Dora Farkas. Their effective leadership took the Math Team to new heights of success. The Stuyvesant Math Team was able to take first place in the NYCIML Junior and Senior Divisions. It was like Alex Khazanov said, "The best thing I like about the math team is how smart the people are." As usual, they also did very well in NYML and At-Pac. There was also a record-breaking 99 persons taking the AIME after getting a score of more than 100 on the AHSME. The most pervading problems in Math Team seemed to be just trying to find out what those abbreviation stood for.

Alarmed and suspicious of this enormous success, the FBI decided to perform an investigation which was thwarted when major witnesses started disappearing. Dora Farkas supposedly fled to Hungary while some other witnesses reportedly got zero-period classes. Felix Liu complained of headaches and switched from a sophomore room to a freshman room. His devoted followers in the sophomore room promptly declared a strike and have been known to run around the room reading newspapers and performing other non-sanctioned activities. Anybody who dared to go in there was met with words like, "Yo, can I borrow the Sports section" and "No man! I have seven more periods."

Most other rooms did not deviate from the daily rituals of math teams, which is coming into the room at 8:00, solving math problems, getting bored, talking, doing some more math, going to sleep, looking out the window, doing more math, doing homework (all subjects except math), going to the bathroom, doing some more math when Mr.Geller (our math team coach) enters the room, etc. But the hot-bed of math action was the senior/junior room of Leo Volftsun. Their devotion to math was demonstrated by their effort in doing a math question for half an hour before Leo announced that it was unsolvable and put up a new question. While others groaned, Michael "I got a perfect score on the AHSME." Develin said, "It's obvious," and solved the question. Meanwhile, Alex Khazanov proved Fermat's last theorem but did not find it necessary to announce such a simple solution to the public (it does not even need a Russian variable) and moved on to other problems. Lyle Goldman looked at the seniors and juniors contemptuously and went to sleep. Mr.Geller meanwhile broke the gang of Rangers trying to predict the future game using probability.

Mr.Geller was happy with the new-found enthusiasm among the members and invited all to come to Bouley's, his favorite restaurant. He even showed the math team the great reviews granted to Bouley's. The math team refused, as they knew Mr.Geller's world-famous dishes still did not take a permanent place in Bouley's menu.

Math Team had a great year and looks forward to doing even better next year. They are just praying that the FBI does not come back this year.



At the Mathematics Competition Resource Center:
IN THE MIDDLE OF NOWHERE

BY DORA FARKAS

LITTLE DID I know what I was getting myself into when I sent in the first set of solutions to the USA Mathematical Talent Search. Little did I know that although I never had an all-nighter at Stuyvesant, I wouldn't be able to survive the following summer without one. Also, least of all did I know that I would meet the great-grandson of Daniel Berzsenyi, one of Hungary's greatest poets. (By the way, I am Hungarian, if you haven't guessed already.)

Although I love writing mathematical proofs, it was especially George Berzsenyi's name on the Talent Search stationery that caught my attention. I had heard him speak once at a math competition at Penn State University. So, I completed all four sets of problems that he posed through the Talent Search and was enchanted by the notice that I was eligible for the Young Scholars program at the Rose-Hulman Institute of Technology, in Indiana, the following summer. Dr. Berzenyi used to write problems for the AHSME, the AIME, and the USAMO, three contests for which people spend years preparing. Because the following year would be my last opportunity to improve my scores, it didn't seem like a bad idea to apply to Rose-Hulman.

And I wasn't at all disappointed. We indeed spent a great deal of the program learning techniques for the contests previously mentioned. Not only that, but we learned Set Theory, Graph Theory, and solved several Kurschak problems, which were originally posed for Hungary's best young mathematicians. We also had access to a NEXT Network and completed many high-quality projects using *Mathematica.*

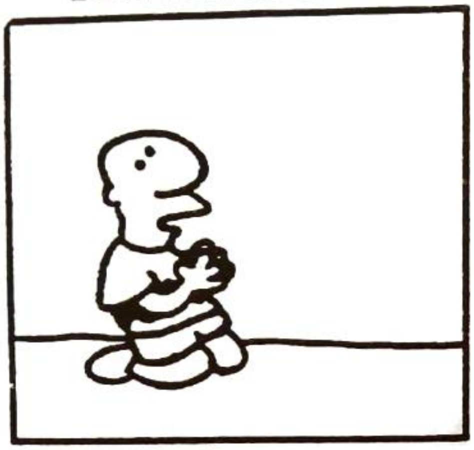
Thanks to a few mischievous young scholars from the Midwest and the South, the program was more adventurous than planned. Playing frisbee with a deck of 52 cards, getting locked into our rooms with the doorknobs outside, waking up at night to mysterious phonecalls, and ordering pizza well after curfew were only a few of the fun things that we enjoyed. Naturally, many "official" recreational activities were organized, including a trip to Columbia House, the Digital Audio Corporation, and a hike at the Turkey Run State Park.

The Young Scholars Program, as well as the Talent Search, has achieved a prestigious status after only its fourth year. All six members of this year's USA Mathematics Team and almost all members of the USA Physics Team were former Talent Search contestants or Young Scholars.

For more information write to:
USAMTS
Department of Mathematics/Box121
Rose-Hulman Institute of Technology
Terre-Haute IN 47803

Mechanics is the paradise of the mathematical sciences because by means of it one comes to the fruits of mathematics.
— Leonardo da Vinci

WHO WOULD YOU RATHER BE, AFTER TAKING THE NEW SAT?



Stan

Joe

Stan and Joe are both intelligent high school students.
Stan and Joe both registered to take the NEW SAT.
Stan felt that he was ready to take the NEW SAT.
Joe knew better. He had heard of The Princeton Review.
Over the past eight years, four independent studies have found The Princeton Review's average score increases to range from 110 to 160 points.
Joe also heard that The Princeton Review closely tracked the changes in the NEW SAT and revamped all of its materials.
Joe thought that would help.
The Princeton Review guarantees its results.
Joe's parents thought that would help.
So Joe sacrificed some hours and took The Princeton Review SAT course.
Stan and Joe both took the NEW SAT.
Both used "number two" pencils.
Both applied to the same colleges.

Who would you rather be?

For course information and a free copy of "The No-Stress Guide to the NEW SAT" call:

THE PRINCETON REVIEW

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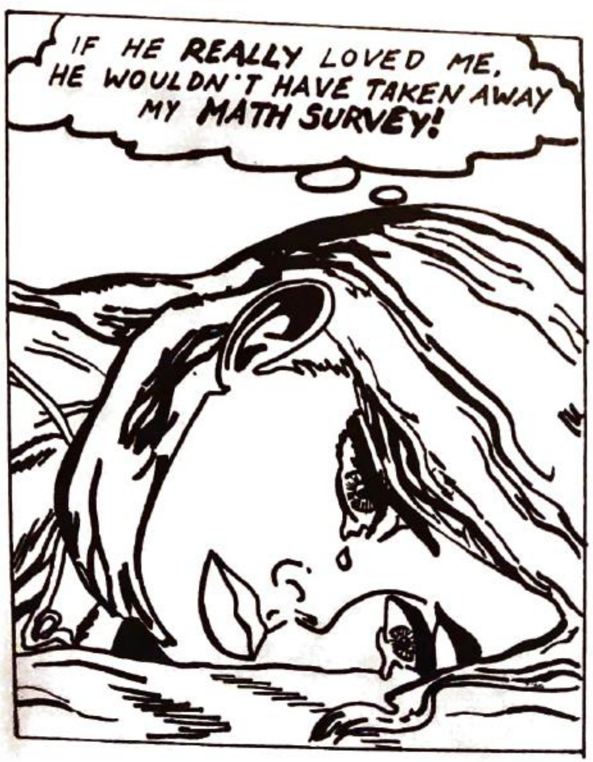

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CONTENTS